

Reference-free logged energy-oracle recovery for neural approximations of symmetric coercive variational problems: conforming Riesz reconstruction and archive-level selection

Karim Bounja^{1,*} Lahcen Laayouni² Boujemaa Achchab¹ Abdeljalil Sakat¹

Abstract

Neural PDE training yields a finite checkpoint archive, yet its logged energy errors are inaccessible without the exact solution, while loss-based selection does not necessarily recover the logged energy oracle. For admissible neural approximations of symmetric coercive variational problems, we introduce a reference-free selection rule based on minimizing a computable conforming Riesz monitor. The exact residual–energy identity and conforming projection make the monitor an unconditional lower bound converging monotonically to each logged energy error under nested conforming refinement; under saturation, hierarchical enrichment yields a computable upper estimate and hence a lower–upper bracket. A key finding is that archive selection is order-sensitive: unresolved checkpoint-dependent components can reverse the oracle–non-oracle ranking at finite resolution, so checkpointwise recovery alone is insufficient. For finite archives, we prove uniform recovery, yielding convergence to the logged-oracle error and, without saturation, logged-oracle selection at sufficiently fine auxiliary resolution. Under saturation, the bracket gives a computable near-oracle bound and certifies unique logged-oracle selection upon interval separation. We also bound logging-resolution loss and certify oracle inclusion over prescribed comparison trajectories. The resulting criterion replaces inaccessible exact-error minimization by computable, training-independent post-training selection on the intrinsic energy-error scale, requiring only the computed candidates and the variational problem. Experiments on diffusion and elasticity, including a non-manufactured perforated plate, demonstrate energy-scale calibration, oracle-level selection, and modest post-processing cost.

Keywords: Reference-free checkpoint selection; Neural PDE solvers; Symmetric coercive variational problems; Conforming Riesz reconstruction; Energy-norm error estimation; Oracle selection consistency.

1 Introduction

Neural solvers are increasingly used to approximate solutions of partial differential equations, yet their practical reliability depends on assessing the computed output in a norm intrinsic to the underlying problem [1, 2]. When training yields a finite archive $\{u_{\theta_k}\}_{k \in \mathcal{K}_{\log}}$, a recorded output is typically retained as the numerical solution. Let $(\mathcal{X}, \|\cdot\|_{\mathcal{X}})$ denote the problem-dependent solution space and the norm in which approximation errors are to be assessed. The reference selection criterion is then to retain the available approximation closest to the exact solution, that is, to identify

$$\arg \min_{k \in \mathcal{K}_{\log}} \|u_{\theta_k} - u^*\|_{\mathcal{X}}.$$

¹Laboratory for Analysis and Modeling of Systems and Decision Support (LAMSD), Hassan 1st University of Settat, Settat 26000, Morocco.

²Department of Computer Science, School of Science and Engineering, Al Akhawayn University in Ifrane, Ifrane 53000, Morocco.

*Corresponding author: k.bounja.doc@uhp.ac.ma.

This ideal error criterion cannot be evaluated directly because the exact solution u^* is unknown. Accordingly, checkpoint selection on the problem-intrinsic error scale calls for computable diagnostics aligned with the same intrinsic geometry.

For the symmetric coercive problems considered here, let $\mathcal{A} := u_b + V_0$ denote the affine admissible class, where V_0 is the space of homogeneous admissible directions. Since the exact solution u^* and every admissible recorded approximation u_{θ_k} belong to $\mathcal{A}$, $u_{\theta_k} - u^* \in V_0$. The symmetric coercive bilinear form a induces the energy norm $\|v\|_a := a(v, v)^{1/2}$ on V_0 . For the finite checkpoint archive, define

$$E_k := \|u_{\theta_k} - u^*\|_a, \quad \mathcal{K}_E^* := \arg \min_{k \in \mathcal{K}_{\log}} E_k.$$

Although the errors E_k are inaccessible, the quadratic Ritz functional $J(u) := \frac{1}{2}a(u, u) - \ell(u)$ and the variational residual

$$\langle R(u), v \rangle := \ell(v) - a(u, v), \quad v \in V_0,$$

satisfy, for every logged checkpoint,

$$J(u_{\theta_k}) - J(u^*) = \frac{1}{2}E_k^2 = \frac{1}{2}\|R(u_{\theta_k})\|_{V_0', a}^2.$$

Thus, the energy-dual residual is exactly calibrated with the target energy error.

For a finite-dimensional conforming auxiliary space $V_m \subset V_0$, let $z_{m,k} \in V_m$ satisfy

$$a(z_{m,k}, v) = \langle R(u_{\theta_k}), v \rangle, \quad v \in V_m,$$

and define the conforming Riesz monitor $\eta_{m,k} := \|z_{m,k}\|_a$. Conforming projection gives

$$\eta_{m,k} \leq E_k, \quad \eta_{m,k} \uparrow E_k$$

along a nested conforming auxiliary hierarchy with dense union in V_0 . We therefore select checkpoints by

$$\mathcal{K}_m^* := \arg \min_{k \in \mathcal{K}_{\log}} \eta_{m,k}.$$

The central difficulty for archive-level selection is that checkpointwise recovery does not by itself preserve the target-error ordering at finite resolution. We identify this ordering obstruction and show that checkpoint-dependent unresolved components may reverse the oracle–non-oracle ranking. For a finite archive, we prove that checkpointwise recovery upgrades to the uniform estimate

$$\varepsilon_m := \max_{k \in \mathcal{K}_{\log}} |E_k - \eta_{m,k}| \longrightarrow 0.$$

This uniform recovery eventually restores the oracle–non-oracle separation. Consequently, every $k_m^* \in \mathcal{K}_m^*$ satisfies

$$0 \leq E_{k_m^*} - E_* \leq \varepsilon_m, \quad E_* := \min_{k \in \mathcal{K}_{\log}} E_k,$$

and, for all sufficiently large m ,

$$\mathcal{K}_m^* \subseteq \mathcal{K}_E^*.$$

Thus, monitor minimization is consistent in value and eventually recovers the logged energy oracle without any saturation assumption.

At finite auxiliary resolution, nested conforming reconstructions provide, under saturation, computable near-oracle bounds and exact logged-oracle certification by interval separation. We further separate auxiliary resolution from logging resolution: relative to a prescribed finite comparison trajectory, competitive-coverage bounds quantify the loss induced by archive subsampling and yield certificates of trajectory-oracle inclusion and unique recovery. The resulting framework replaces inaccessible exact-error minimization by a computable, training-independent post-training selection procedure requiring neither the exact solution nor a reference solve.

The present work lies at the intersection of two lines of analysis: error estimates relating residual or variational training objectives to solution errors under suitable stability assumptions [2, 3], and a posteriori constructions providing computable error indicators for neural PDE approximations [4, 5].

For training objectives intended to provide quantitative control of the error with respect to the exact solution, recent work has identified calibration with the problem-intrinsic error as a structural design requirement. Consistent collocation losses providing error control in the target energy norm have been constructed [6], while variationally correct residual losses whose values are uniformly equivalent to the squared solution error in the norm determined by a stable variational formulation have been developed in [7]. Related strategies have also incorporated computable residual-based error information directly into the training objective, including discrete dual residual norms and a posteriori estimators [8, 9].

For archive selection, however, even uniform checkpointwise loss–error equivalence does not by itself guarantee preservation of a minimum-error checkpoint under loss minimization over a finite archive. Indeed, suppose that, for some $p > 0$ and constants $0 < c < C$,

$$cE_k^p \leq \mathcal{L}_k \leq CE_k^p, \quad k \in \mathcal{K}_{\log}.$$

If two checkpoints satisfy $E_i < E_j$ while $CE_i^p > cE_j^p$, these bounds are compatible with the admissible values $\mathcal{L}_i = CE_i^p$ and $\mathcal{L}_j = cE_j^p$, for which $\mathcal{L}_i > \mathcal{L}_j$. Thus, uniform loss–error equivalence controls each candidate checkpointwise but does not, by itself, imply preservation of the target-error ordering or guarantee that every loss minimizer over the archive is a target-error minimizer.

Independent post-training error assessment offers a complementary route: rather than inferring target accuracy from the training loss, a posteriori approaches provide computable information on the error of a computed approximation. Representative constructions include a dual-weighted-residual estimator for Deep Ritz approximations of the Laplace and Stokes problems [4], a reliable and efficient energy-norm estimator for VPINNs combining residual, loss, and data-oscillation contributions [5], and rigorous lower and upper error bounds obtained from Riesz representations of residual extensions and restrictions to geometrically simpler domains [10]. A related line constructs neural approximation spaces sequentially, with the resulting updates providing error indicators and stopping criteria in Galerkin and collocation settings [11, 12]; an extended Galerkin framework addresses general boundary-value problems with singular solution structures [13]. For general well-posed variational formulations, checkpointwise error control has been obtained through a computable projected-residual contribution together with a complementary residual contribution admitting reliable a posteriori control [9]. These approaches provide checkpointwise error assessment. The finite-resolution ordering obstruction for the conforming Riesz monitor, analyzed in Proposition 8, however, shows that checkpointwise control alone does not ensure reliable archive-level selection. More generally, target-error order preservation is not guaranteed beyond this setting; see Section 7.

The present work therefore makes finite-archive checkpoint selection itself the object of analysis in the symmetric coercive setting, identifying the finite-resolution ordering obstruction and establishing its removal through uniform recovery over the archive. The resulting reference-free rule is

$$\mathcal{K}_m^* := \arg \min_{k \in \mathcal{K}_{\log}} \eta_{m,k}.$$

For finite archives, conforming auxiliary refinement yields eventual logged-oracle selection; under saturation, nested reconstructions provide finite-resolution near-oracle bounds and interval-separation certificates, while logging-resolution control quantifies the loss relative to a prescribed finite comparison trajectory and yields trajectory-oracle certificates. Thus, once optimization has generated sufficiently accurate admissible candidates—the exact Ritz objective being energy-calibrated by Lemma 1—monitor refinement recovers the logged oracle, while logging-resolution control determines whether it is also an oracle of the optimization trajectory.

To the best of our knowledge, this passage from checkpointwise error assessment to reference-free archive-level oracle selection has not previously been analyzed for neural PDE approximations. The framework enables reference-free recovery and certification of energy-optimal checkpoints within the retained archive and relative to a prescribed finite comparison trajectory.

The numerical study tests calibration, auxiliary and logging refinement, and oracle selection on manufactured diffusion and elasticity benchmarks and on a non-manufactured perforated plate assessed against

an independent refined FEM reference. The results show energy-error calibration at sufficient auxiliary resolution, oracle-level selection, controlled logging loss, and low monitoring overhead.

The remainder of the paper is organized as follows. Sections 2–5 develop the variational setting, conforming Riesz reconstruction, logged-oracle guarantees, and logging-resolution control. Section 6 presents the numerical validation, and Section 7 discusses broader implications for archive-level selection.

2 Problem setting

We consider symmetric coercive variational problems posed on an affine admissible class and approximated by parametrized functions belonging to that same class.

2.1 Abstract symmetric coercive setting

Let X be a real Hilbert space and $V_0 \subset X$ a closed linear subspace representing the homogeneous admissible directions associated with the essential constraints. Given a lifting $u_b \in X$ of the prescribed essential data, define

$$\mathcal{A} := u_b + V_0.$$

Thus, admissible functions belong to $\mathcal{A}$, while differences of admissible functions belong to V_0 . For homogeneous essential data, $u_b = 0$ and $\mathcal{A} = V_0$.

Let $a : X \times X \rightarrow \mathbb{R}$ be a continuous symmetric bilinear form whose restriction to $V_0 \times V_0$ is coercive. Thus, there exist constants $0 < \alpha_a \leq \beta_a < \infty$ such that

$$\begin{aligned} |a(w, v)| &\leq \beta_a \|w\|_X \|v\|_X, & \forall w, v \in X, \\ a(v, v) &\geq \alpha_a \|v\|_X^2, & \forall v \in V_0. \end{aligned}$$

Let $\ell \in X'$. The exact solution is the unique $u^* \in \mathcal{A}$ satisfying

$$a(u^*, v) = \ell(v), \quad \forall v \in V_0. \tag{1}$$

Indeed, subtracting the lifting u_b reduces the affine problem to a coercive problem on V_0 , to which the Lax–Milgram theorem applies.

The bilinear form a induces the energy inner product and norm on V_0 ,

$$(w, v)_a := a(w, v), \quad \|v\|_a^2 := a(v, v), \quad w, v \in V_0.$$

For $u \in X$, define the Ritz functional $J(u) := \frac{1}{2}a(u, u) - \ell(u)$. Its minimization over $\mathcal{A}$ has (1) as its first-order optimality condition.

Let V_0' denote the topological dual of V_0 . For every $u \in \mathcal{A}$, the map

$$v \mapsto \ell(v) - a(u, v)$$

is continuous on V_0 . We therefore define the variational residual $R(u) \in V_0'$ by

$$\langle R(u), v \rangle := \ell(v) - a(u, v), \quad \forall v \in V_0,$$

and measure it in the dual norm induced by the energy norm,

$$\|F\|_{V_{0,a}'} := \sup_{v \in V_0 \setminus \{0\}} \frac{|\langle F, v \rangle|}{\|v\|_a}, \quad F \in V_0'.$$

The residual–energy identities of Section 3 are formulated on V_0 , while the Riesz reconstructions and hierarchical estimators of Subsection 3.3 are defined on conforming subspaces of V_0 . The key admissibility property is

$$u - u^* \in V_0, \quad u, u^* \in \mathcal{A},$$

which allows the energy-dual norm of the variational residual to be identified with the energy error.

2.2 Admissible neural Ritz approximations

A parametrized approximation u_θ is called admissible if $u_\theta \in \mathcal{A}$, and therefore $u_\theta - u^* \in V_0$, as required by the residual–energy identity.

For concrete boundary-value problems, let Γ_D denote the portion of the boundary carrying Dirichlet-type essential conditions. Admissibility may be enforced through a hard boundary ansatz chosen so that its homogeneous correction belongs to V_0 . For homogeneous essential data, one may use

$$u_\theta(x) = g(x)N_\theta(x), \quad g = 0 \quad \text{on } \Gamma_D, \quad (2)$$

where N_θ is the free neural network and the boundary factor g is chosen so that $gN_\theta \in V_0$. For nonhomogeneous essential data, one may use

$$u_\theta(x) = u_b(x) + g(x)N_\theta(x), \quad g = 0 \quad \text{on } \Gamma_D. \quad (3)$$

The first construction yields $u_\theta = 0$ on Γ_D , while the second yields $u_\theta = u_b$ there; with the stated choice of boundary factor, both place u_θ in $\mathcal{A}$.

If the neural trial class is not constrained to lie in $\mathcal{A}$ and essential boundary conditions are imposed only through a finite-weight penalty in the training objective, admissibility is not guaranteed. Although the penalty may reduce essential-boundary violations, it does not in general enforce the essential boundary condition exactly. Consequently, $u_\theta - u^*$ is not guaranteed to belong to V_0 , and the residual–energy identity (5) does not apply directly to the original constrained problem without an admissible correction or a modified formulation. This contrasts with (2)–(3), which enforce admissibility by construction. Natural boundary conditions, such as Neumann data and symmetric Robin terms, instead enter the variational formulation through $\ell(\cdot)$, or through $a(\cdot, \cdot)$ and $\ell(\cdot)$, rather than defining additional affine constraints on u_θ .

3 Residual–energy structure and conforming Riesz error assessment

Within the symmetric coercive setting of Section 2, this section develops the residual–energy structure and its computable conforming realization for admissible neural approximations. The analysis combines the quadratic Ritz identity, Fréchet–Riesz representation, conforming Galerkin projection, and nested-space estimation under saturation [14–18]. The resulting lower monitor η_m and, under saturation, the hierarchical bracket (22) provide the analytical basis for the checkpoint-selection and oracle guarantees developed in Section 4.

3.1 Exact residual–energy identities

We first record the exact identities linking the Ritz functional, the energy-dual residual, and the energy error. For admissible $u \in \mathcal{A}$, these identities identify $\|R(u)\|_{V'_{0,a}}$ as the residual quantity exactly calibrated with $\|u - u^*\|_a$.

Lemma 1 (Energy-gap identity). *For every $u \in \mathcal{A}$,*

$$J(u) - J(u^*) = \frac{1}{2} \|u - u^*\|_a^2.$$

Proof. Set $e := u - u^* \in V_0$. Using $u = u^* + e$, symmetry of a , and $\ell(e) = a(u^*, e)$, we obtain

$$\begin{aligned} J(u) - J(u^*) &= a(u^*, e) + \frac{1}{2} a(e, e) - \ell(e) \\ &= \frac{1}{2} a(e, e) = \frac{1}{2} \|e\|_a^2. \end{aligned}$$

□

Proposition 1 (Exact residual–energy identity). *For every $u \in \mathcal{A}$,*

$$\|R(u)\|_{V'_{0,a}} = \|u - u^*\|_a. \quad (4)$$

Consequently,

$$J(u) - J(u^*) = \frac{1}{2} \|R(u)\|_{V'_{0,a}}^2 = \frac{1}{2} \|u - u^*\|_a^2. \quad (5)$$

Proof. Set $e := u - u^* \in V_0$. For every $v \in V_0$,

$$\langle R(u), v \rangle = \ell(v) - a(u, v) = -a(e, v).$$

Hence, by Cauchy–Schwarz,

$$\|R(u)\|_{V_{0,a}'} \leq \|e\|_a.$$

If $e \neq 0$, choosing $v = -e$ in the definition of the dual norm gives the reverse inequality. The case $e = 0$ is immediate, and (5) follows from Lemma 1. $\square$

Thus, for admissible approximations, the energy-dual residual norm coincides exactly with the energy error. Minimizing $\|R(u)\|_{V_{0,a}'}$ over admissible candidates is therefore equivalent to minimizing the unknown energy error, independently of how the candidates were generated. This identifies the error-calibrated quantity targeted by the conforming Riesz reconstruction.

Remark 1 (Ritz ordering and error quantification). Identity (5) also shows that exact Ritz values preserve the ordering of the energy errors. They do not, however, quantify the energy error without the unknown reference level $J(u^*)$. The energy-dual residual norm removes this reference dependence: it equals the energy error and admits the computable conforming realization introduced in Subsection 3.3.

Remark 2 (Strong residuals as indirect proxies). Suppose that the residual of an admissible approximation u admits an L^2 -representation

$$\langle R(u), v \rangle = - \int_{\Omega} r_s(u) \cdot v \, dx, \quad \forall v \in V_0,$$

and that, for some $C_{\text{emb}} > 0$,

$$\|v\|_{L^2(\Omega)} \leq C_{\text{emb}} \|v\|_a, \quad \forall v \in V_0.$$

Then

$$\|R(u)\|_{V_{0,a}'} \leq C_{\text{emb}} \|r_s(u)\|_{L^2(\Omega)}.$$

In standard elliptic settings, such a bound follows from the relevant Poincaré–Friedrichs, Korn, or norm-equivalence estimates; see, e.g., [14, 15]. Thus, strong-form L^2 residuals may control the energy-dual residual norm through a problem-dependent constant, but they remain indirect proxies rather than the exactly calibrated residual quantity.

3.2 Variational consequences for admissible trial sets

We first record a direct consequence of the exact Ritz–energy identity for admissible trial sets.

Proposition 2 (Best energy approximation over arbitrary admissible trial sets). *Let $\Sigma \subset \mathcal{A}$ be any nonempty admissible trial set, not necessarily linear or convex, and assume that*

$$u_{\Sigma} \in \arg \min_{w \in \Sigma} J(w).$$

Then

$$\|u_{\Sigma} - u^*\|_a = \inf_{w \in \Sigma} \|w - u^*\|_a.$$

Proof. Since $\Sigma \subset \mathcal{A}$, every $w \in \Sigma$ satisfies $w - u^* \in V_0$. By Lemma 1,

$$J(w) - J(u^*) = \frac{1}{2} \|w - u^*\|_a^2, \quad w \in \Sigma.$$

Since $J(u^*)$ is independent of w , minimizing J over Σ is equivalent to minimizing $\|w - u^*\|_a$ over Σ . $\square$

Proposition 2 identifies the exact continuous optimization geometry underlying admissible neural Ritz approximation: any exact minimizer of the Ritz functional over an admissible trial set is an energy-best element of that set, namely, a closest element to the exact solution in the energy norm.

This exact statement concerns the continuous functional J . If J is replaced by an empirical or quadrature-based approximation $\tilde{J}$, the exact ordering is not automatically preserved. Indeed, for admissible candidates u_i, u_j ,

$$J(u_i) < J(u_j) \iff \|u_i - u^*\|_a < \|u_j - u^*\|_a.$$

If

$$|\tilde{J}(u_k) - J(u_k)| \leq \varepsilon_k, \quad k \in \{i, j\},$$

then this ordering is guaranteed to be preserved whenever

$$J(u_j) - J(u_i) > \varepsilon_i + \varepsilon_j.$$

When this separation condition is not available, the approximation bounds alone do not guarantee preservation of the energy-error ordering. Consequently, a certified connection between the practical training problem and exact Ritz minimization requires quantitative control of both the functional-approximation error and the optimization error.

This motivates a post-training assessment targeted directly at the energy error. By Proposition 1, the energy-dual residual norm provides an exactly calibrated target without access to u^* ; the next subsection constructs its computable conforming approximation.

Remark 3 (Relation to Ritz–Galerkin best approximation). For $\Sigma = u_b + W_h$, with $W_h \subset V_0$ a conforming linear subspace, Proposition 2 reduces to the classical Ritz–Galerkin best-approximation property [14, 15]. Here the same conclusion holds for any nonempty admissible $\Sigma \subset \mathcal{A}$, without linearity or convexity assumptions.

Proposition 3 (Near-minimizer control). *Let $u \in \mathcal{A}$ and $\varepsilon \geq 0$. If*

$$J(u) - J(u^*) \leq \varepsilon,$$

then

$$\|u - u^*\|_a \leq \sqrt{2\varepsilon}, \quad \|R(u)\|_{V'_{0,a}} \leq \sqrt{2\varepsilon}.$$

Proof. The result follows directly from Lemma 1 and Proposition 1. □

Corollary 1 (Residual convergence and energy-error control). *Let $(u_n)_n \subset \mathcal{A}$. Then:*

(i) *if $\|R(u_n)\|_{V'_{0,a}} \rightarrow 0$, then $u_n \rightarrow u^*$ in the energy norm;*

(ii) *if $u \in \mathcal{A}$ and $\|R(u)\|_{V'_{0,a}} \leq \varepsilon$, then $\|u - u^*\|_a \leq \varepsilon$.*

Proof. Both statements follow directly from Proposition 1. □

3.3 Conforming Riesz reconstruction and hierarchical error bounds

We now turn the exact residual–energy relation into a computable post-training assessment. By Corollary 1, reducing $\|R(u_\theta)\|_{V'_{0,a}}$ drives the energy error to zero, while any bound on this residual norm transfers directly to $\|u_\theta - u^*\|_a$. We therefore approximate the Riesz representative of $R(u_\theta)$ in conforming auxiliary subspaces of V_0 . A single auxiliary space yields a lower monitor of the resolved energy-error component, while a nested pair quantifies the component revealed by enrichment and, under saturation, provides a conditional upper estimate.

3.3.1 Continuous Riesz representative

For an admissible neural Ritz approximation $u_\theta \in \mathcal{A}$, the residual $R(u_\theta)$ belongs to V'_0 . By the Riesz representation theorem in the Hilbert space $(V_0, \|\cdot\|_a)$, there exists a unique $z \in V_0$ such that

$$a(z, v) = \langle R(u_\theta), v \rangle, \quad \forall v \in V_0. \tag{6}$$

Proposition 4 (Continuous Riesz representative as the signed error). *Let $u_\theta \in \mathcal{A}$, and let $z \in V_0$ solve (6). Then*

$$z = u^* - u_\theta,$$

and consequently

$$\|z\|_a = \|u_\theta - u^*\|_a = \|R(u_\theta)\|_{V'_{0,a}}. \tag{7}$$

Proof. Since $u_\theta, u^* \in \mathcal{A}$, $u^* - u_\theta \in V_0$. For every $v \in V_0$,

$$a(z, v) = \langle R(u_\theta), v \rangle = \ell(v) - a(u_\theta, v) = a(u^* - u_\theta, v).$$

Uniqueness of the Riesz representative gives $z = u^* - u_\theta$. The norm identities follow from Proposition 1. $\square$

Thus, the exact signed error is the continuous Riesz representative of the residual. The computational problem is therefore to approximate this representative in a conforming auxiliary space.

3.3.2 Single-space conforming realization

Let $V_m \subset V_0$ be a finite-dimensional conforming auxiliary subspace. The discrete Riesz representative $z_m \in V_m$ is defined by

$$a(z_m, v_m) = \langle R(u_\theta), v_m \rangle, \quad \forall v_m \in V_m. \quad (8)$$

We define

$$\eta_m(u_\theta) := \|z_m\|_a. \quad (9)$$

For clarity, the auxiliary problem approximates the Riesz representative of the residual generated by u_θ ; it is not a Galerkin approximation of the primal solution u^* .

Since $V_m \subset V_0$, problem (8) is the conforming Galerkin approximation of the continuous Riesz problem (6). Galerkin orthogonality identifies z_m as the a -orthogonal projection of z onto V_m [14, 15]. In the present residual–Riesz setting, this projection yields the computable lower-monitor properties stated below.

Proposition 5 (Conforming projection and lower monitor). *Let $z \in V_0$ and $z_m \in V_m$ solve (6) and (8), respectively. Then*

$$a(z - z_m, v_m) = 0, \quad \forall v_m \in V_m, \quad (10)$$

so that z_m is the best approximation of z in V_m :

$$\|z - z_m\|_a = \inf_{v_m \in V_m} \|z - v_m\|_a.$$

Moreover,

$$\|u_\theta - u^*\|_a^2 = \eta_m(u_\theta)^2 + \|z - z_m\|_a^2, \quad (11)$$

and therefore

$$\eta_m(u_\theta) \leq \|u_\theta - u^*\|_a = \|R(u_\theta)\|_{V'_{0,a}}. \quad (12)$$

Proof. Subtracting (8) from (6) gives (10). Hence z_m is the a -orthogonal projection of z onto V_m , so

$$\|z\|_a^2 = \|z_m\|_a^2 + \|z - z_m\|_a^2.$$

Using Proposition 4 and $\eta_m(u_\theta) = \|z_m\|_a$ yields (11) and (12). $\square$

Thus, solving the finite-dimensional auxiliary problem (8) yields $\eta_m(u_\theta)$ as a computable conforming lower monitor for the energy error, while $\|z - z_m\|_a$ is the component unresolved by V_m . This decomposition motivates nested auxiliary spaces, which quantify the additional component revealed by enrichment.

Corollary 2 (Monotone recovery under conforming refinement). *Let $(V_m)_{m \geq 1}$ be a nested sequence of finite-dimensional conforming subspaces such that*

$$V_m \subset V_{m+1} \subset V_0, \quad \overline{\bigcup_{m \geq 1} V_m}^{\|\cdot\|_a} = V_0.$$

For a fixed $u_\theta \in \mathcal{A}$, let $z_m \in V_m$ solve (8). Then $\eta_m(u_\theta) = \|z_m\|_a$ is nondecreasing and

$$\eta_m(u_\theta) \uparrow \|u_\theta - u^*\|_a = \|R(u_\theta)\|_{V'_{0,a}}. \quad (13)$$

Proof. By Proposition 5,

$$\eta_m(u_\theta)^2 = \|z\|_a^2 - \|z - z_m\|_a^2.$$

Nestedness makes $\|z - z_m\|_a$ nonincreasing, while density and the best-approximation property imply $\|z - z_m\|_a \rightarrow 0$. Hence $\eta_m(u_\theta)$ increases to $\|z\|_a$, yielding (13). $\square$

Thus, $\eta_m(u_\theta)$ recovers the exact energy error monotonically from below under conforming refinement. Over a finite checkpoint archive, this property provides the basis for the reference-free checkpoint-selection rule and oracle guarantees developed in Section 4.

A finite enrichment $V_m \subset V_M$ is considered next to quantify the additional error component resolved beyond V_m .

3.3.3 Nested-space enrichment and hierarchical gap

Let $V_m \subset V_M \subset V_0$ be two finite-dimensional conforming auxiliary subspaces. In addition to the reconstruction $z_m \in V_m$, define the enriched Riesz reconstruction $z_M \in V_M$ by

$$a(z_M, v_M) = \langle R(u_\theta), v_M \rangle, \quad \forall v_M \in V_M. \quad (14)$$

We set

$$\eta_M(u_\theta) := \|z_M\|_a$$

and define the hierarchical gap

$$\delta_{m,M}(u_\theta) := (\eta_M(u_\theta)^2 - \eta_m(u_\theta)^2)^{1/2}. \quad (15)$$

Proposition 6 (Nested-space orthogonal decomposition). *Let $u_\theta \in \mathcal{A}$, let $z \in V_0$ solve (6), and let $z_m \in V_m$ and $z_M \in V_M$ solve (8) and (14), respectively. If $V_m \subset V_M \subset V_0$, then*

$$\eta_m(u_\theta)^2 \leq \eta_M(u_\theta)^2 \leq \|u_\theta - u^*\|_a^2.$$

Moreover,

$$\delta_{m,M}(u_\theta)^2 = \eta_M(u_\theta)^2 - \eta_m(u_\theta)^2 = \|z_M - z_m\|_a^2, \quad (16)$$

and the exact energy error admits the orthogonal decomposition

$$\|u_\theta - u^*\|_a^2 = \eta_m(u_\theta)^2 + \delta_{m,M}(u_\theta)^2 + \|z - z_M\|_a^2. \quad (17)$$

Proof. Since $V_m \subset V_M$, subtracting the two discrete Riesz problems gives

$$a(z_M - z_m, v_m) = 0, \quad \forall v_m \in V_m.$$

Hence

$$\|z_M\|_a^2 = \|z_m\|_a^2 + \|z_M - z_m\|_a^2,$$

which yields (16) and $\eta_m(u_\theta) \leq \eta_M(u_\theta)$. Moreover, Galerkin orthogonality at level M gives

$$\|z\|_a^2 = \|z_M\|_a^2 + \|z - z_M\|_a^2.$$

Combining the two identities and using $z = u^* - u_\theta$ yields (17) and the remaining inequality. $\square$

Remark 4 (Hierarchical interpretation). The decomposition in Proposition 6 is the nested-projection geometry underlying hierarchical a posteriori estimation [16, 17, 19]. Here the projected quantity is the signed energy error: $\eta_m(u_\theta)$ measures the component resolved in V_m , $\delta_{m,M}(u_\theta)$ the additional component revealed by enrichment to V_M , and $\|z - z_M\|_a$ the remaining unresolved component.

Because the auxiliary hierarchy is independent of the network parametrization and training procedure, the same construction can be applied post-training to a finite checkpoint archive. Under saturation, $\eta_m(u_\theta)$ and $\delta_{m,M}(u_\theta)$ then provide the lower-upper information used for the finite-level selection guarantees of Section 4.

To control the unresolved component $\|z - z_M\|_a$ through the computable gap $\delta_{m,M}(u_\theta)$, nestedness alone is insufficient. We therefore introduce the standard saturation condition from hierarchical a posteriori estimation [16, 17, 19]. It enters only the conditional upper estimate; the lower monitor and its monotone recovery remain unconditional.

Assumption 1 (Saturation). Let $u_\theta \in \mathcal{A}$ be fixed, with continuous Riesz representative $z = u^* - u_\theta$. A nested pair $V_m \subset V_M \subset V_0$ satisfies saturation for u_θ , with factor $q \in (0, 1)$, if

$$\|z - z_M\|_a \leq q \|z - z_m\|_a. \quad (18)$$

Proposition 7 (Conditional upper estimate under saturation). Let $u_\theta \in \mathcal{A}$, and suppose that the nested pair $V_m \subset V_M \subset V_0$ satisfies Assumption 1 for u_θ , with factor $q \in (0, 1)$. Then

$$\|u_\theta - u^*\|_a^2 \leq \eta_m(u_\theta)^2 + \frac{\delta_{m,M}(u_\theta)^2}{1 - q^2}. \quad (19)$$

Accordingly, defining

$$\mathcal{U}_{m,M,q}(u_\theta) := \left(\eta_m(u_\theta)^2 + \frac{\delta_{m,M}(u_\theta)^2}{1 - q^2} \right)^{1/2}, \quad (20)$$

one has

$$\|u_\theta - u^*\|_a \leq \mathcal{U}_{m,M,q}(u_\theta). \quad (21)$$

Proof. By Proposition 6,

$$\delta_{m,M}(u_\theta)^2 = \|z - z_m\|_a^2 - \|z - z_M\|_a^2.$$

Assumption 1 therefore gives

$$\delta_{m,M}(u_\theta)^2 \geq (1 - q^2) \|z - z_m\|_a^2,$$

and hence

$$\|z - z_m\|_a^2 \leq \frac{\delta_{m,M}(u_\theta)^2}{1 - q^2}.$$

Combining this with

$$\|u_\theta - u^*\|_a^2 = \eta_m(u_\theta)^2 + \|z - z_m\|_a^2$$

yields (19) and (21). $\square$

Combining Proposition 5 with Proposition 7 yields the lower–upper bracket

$$\eta_m(u_\theta) \leq \|u_\theta - u^*\|_a \leq \mathcal{U}_{m,M,q}(u_\theta). \quad (22)$$

The lower bound is unconditional, whereas the upper bound at a given checkpoint requires Assumption 1 for that checkpoint and the prescribed factor q . In the finite-archive guarantees developed in Section 4, saturation is required only where the corresponding upper estimate is invoked.

Remark 5 (Dependence on enrichment and eventual saturation). Nestedness alone does not guarantee saturation for an arbitrary finite pair. For fixed V_m , however, let $(V_M)_{M \geq m}$ be a nested enrichment satisfying

$$\overline{\bigcup_{M \geq m} V_M}^{\|\cdot\|_a} = V_0.$$

For each fixed checkpoint k ,

$$\|z_k - z_{M,k}\|_a \longrightarrow 0,$$

while $\|z_k - z_{m,k}\|_a$ is fixed. Hence, for every prescribed $q \in (0, 1)$, saturation holds for all sufficiently large M ; if the coarse projection error vanishes, it holds trivially for every $M \geq m$. For a finite checkpoint archive, the corresponding thresholds can be maximized to obtain a common enrichment level. This argument does not provide a computable threshold, so the upper estimate for a prescribed finite pair remains conditional.

Algorithm 1 Evaluation of the conforming Riesz monitor and hierarchical estimator

Require: Admissible approximation $u_\theta \in \mathcal{A}$; preassembled auxiliary matrices G_m and G_M ; data for assembling the residual vectors on V_m and V_M ; optional prescribed factor $q \in (0, 1)$.

Ensure: Lower monitor η_m , enriched monitor η_M , hierarchical gap $\delta_{m,M}$, and, when q is prescribed, conditional upper quantity $\mathcal{U}_{m,M,q}$.

- 1: **for** $h \in \{m, M\}$ **do**
- 2: Assemble the checkpoint-dependent residual vector r_h .
- 3: Solve $G_h c^{(h)} = r_h$.
- 4: Compute

$$\eta_h(u_\theta) = (r_h^\top c^{(h)})^{1/2}.$$

- 5: **end for**
- 6: Compute

$$\delta_{m,M}(u_\theta) = (\eta_M(u_\theta)^2 - \eta_m(u_\theta)^2)^{1/2}.$$

- 7: **if** $q \in (0, 1)$ is prescribed **then**
- 8: Compute

$$\mathcal{U}_{m,M,q}(u_\theta) = \left(\eta_m(u_\theta)^2 + \frac{\delta_{m,M}(u_\theta)^2}{1 - q^2} \right)^{1/2}.$$

- 9: **end if**

- 10: **return** $\eta_m(u_\theta)$, $\eta_M(u_\theta)$, $\delta_{m,M}(u_\theta)$, and, when computed, $\mathcal{U}_{m,M,q}(u_\theta)$.
-

Operational interpretation of the hierarchical bracket Under the saturation condition, the bracket implies

$$0 \leq \|u_\theta - u^*\|_a - \eta_m(u_\theta) \leq \mathcal{U}_{m,M,q}(u_\theta) - \eta_m(u_\theta).$$

Thus, its width bounds the possible underestimation of the energy error by the lower monitor. A wide bracket indicates that further auxiliary refinement may be required. Applied over a finite checkpoint archive, these intervals provide the finite-level information used in Section 4.

3.3.4 Matrix realization

For $h \in \{m, M\}$, let $\{\phi_i^{(h)}\}_{i=1}^{N_h}$ be a basis of V_h , and write

$$z_h = \sum_{j=1}^{N_h} c_j^{(h)} \phi_j^{(h)}.$$

Define the auxiliary stiffness matrix $G_h \in \mathbb{R}^{N_h \times N_h}$ and the checkpoint-dependent residual vector $r_h \in \mathbb{R}^{N_h}$ by

$$(G_h)_{ij} = a(\phi_j^{(h)}, \phi_i^{(h)}), \quad (r_h)_i = \ell(\phi_i^{(h)}) - a(u_\theta, \phi_i^{(h)}).$$

The discrete Riesz problem is then equivalent to

$$G_h c^{(h)} = r_h. \tag{23}$$

By symmetry and coercivity of a , G_h is symmetric positive definite. Once (23) has been solved, the monitor is obtained from

$$\eta_h(u_\theta) = \|z_h\|_a = \left((c^{(h)})^\top G_h c^{(h)} \right)^{1/2} = \left(r_h^\top c^{(h)} \right)^{1/2}. \tag{24}$$

The matrices G_m and G_M depend only on the bilinear form and the auxiliary spaces and may therefore be preassembled and factorized once. For each checkpoint, only the checkpoint-dependent residual vectors are assembled and the corresponding linear systems solved. Applying the construction at $h = m$ and $h = M$ yields $\eta_m(u_\theta)$ and $\eta_M(u_\theta)$, from which $\delta_{m,M}(u_\theta)$ and, for a prescribed $q \in (0, 1)$, $\mathcal{U}_{m,M,q}(u_\theta)$ follow as summarized in Algorithm 1.

The identities above refer to the exact auxiliary variational forms; their numerical realization under quadrature is assessed separately in Section 6 and C.

4 Reference-free checkpoint selection and logged-oracle guarantees

This section develops the reference-free selection rule by minimizing the conforming Riesz monitor over a finite archive. We identify the finite-resolution ordering obstruction, show that conforming refinement removes it uniformly and yields eventual logged-oracle selection, and derive, under saturation, a computable near-oracle bound and a unique-oracle certificate.

4.1 Monitor minimization and finite-resolution ordering obstruction

Let $\mathcal{K}_{\log}$ denote the finite set of recorded checkpoints and, for brevity, write $u_k := u_{\theta_k}$. Define

$$E_k := \|u^* - u_k\|_a, \quad k \in \mathcal{K}_{\log}.$$

By Proposition 5 and Corollary 2, for every fixed $k \in \mathcal{K}_{\log}$,

$$\eta_{m,k} \leq E_k, \quad \eta_{m,k} \longrightarrow E_k \quad \text{as } m \rightarrow \infty. \quad (25)$$

These checkpointwise properties motivate the computable selection rule

$$k_m^* \in \arg \min_{k \in \mathcal{K}_{\log}} \eta_{m,k}.$$

The corresponding consistency question is whether, for all sufficiently fine auxiliary resolutions,

$$\arg \min_{k \in \mathcal{K}_{\log}} \eta_{m,k} \subseteq \arg \min_{k \in \mathcal{K}_{\log}} E_k.$$

At a fixed auxiliary level, checkpointwise recovery alone does not guarantee this ordering. Let i be an energy-oracle checkpoint and j a non-oracle checkpoint, so that $E_i < E_j$. Although both monitors are valid lower bounds, one may still have

$$\eta_{m,j} < \eta_{m,i},$$

because the unresolved error component may differ across checkpoints.

To characterize this obstruction, define

$$z_k := u^* - u_k, \quad \rho_{m,k} := \|(I - P_m)z_k\|_a,$$

where P_m denotes the a -orthogonal projection onto V_m . Then

$$E_k^2 = \eta_{m,k}^2 + \rho_{m,k}^2.$$

Proposition 8 (Finite-resolution ordering obstruction). *For any two checkpoints $i, j \in \mathcal{K}_{\log}$,*

$$\eta_{m,j}^2 - \eta_{m,i}^2 = (E_j^2 - E_i^2) - (\rho_{m,j}^2 - \rho_{m,i}^2).$$

Consequently, if $E_i < E_j$, then the monitor ordering is reversed,

$$\eta_{m,i} > \eta_{m,j},$$

if and only if

$$\rho_{m,j}^2 - \rho_{m,i}^2 > E_j^2 - E_i^2.$$

Moreover, if

$$\{0\} \subsetneq V_m \subsetneq V_0,$$

then such a reversal is compatible with the present assumptions: there exist admissible approximations $u_i, u_j \in \mathcal{A}$ such that

$$E_i < E_j \quad \text{and} \quad 0 < \eta_{m,j} < \eta_{m,i}.$$

Proof. For each checkpoint k , the a -orthogonal decomposition

$$z_k = P_m z_k + (I - P_m) z_k$$

gives

$$\eta_{m,k}^2 = E_k^2 - \rho_{m,k}^2.$$

Subtracting the identities for i and j yields the first claim and, when $E_i < E_j$, the stated characterization of rank reversal.

To show that such a reversal can occur, choose a -unit vectors

$$x \in V_m, \quad y \in V_m^{\perp a} \cap V_0,$$

and let $0 < \gamma < \alpha < \beta$. Define

$$z_i := \alpha x, \quad z_j := \gamma x + \beta y, \quad u_i := u^* - z_i, \quad u_j := u^* - z_j.$$

Then $u_i, u_j \in \mathcal{A}$, and

$$E_i = \alpha < \sqrt{\gamma^2 + \beta^2} = E_j.$$

Since

$$P_m z_i = \alpha x, \quad P_m z_j = \gamma x,$$

one has

$$\eta_{m,i} = \alpha > \gamma = \eta_{m,j} > 0.$$

□

Thus, the finite-resolution obstruction is caused by checkpoint-dependent projection defects, not by failure of the lower-bound property itself. At fixed V_m , the unresolved component $\rho_{m,k}$ depends on the direction of the error z_k relative to the auxiliary space, not only on its norm. Different checkpoints may therefore have different resolved and unresolved error components, allowing the unresolved-component difference to overcome the exact energy gap and reverse the ordering.

4.2 Logged-oracle consistency under conforming refinement

We now show that the finite-resolution obstruction identified in Proposition 8 cannot persist under a common nested conforming refinement of a finite checkpoint archive. The key point is that checkpointwise recovery becomes uniform over finitely many checkpoints.

Define the logged energy-oracle level and oracle set by

$$E_* := \min_{k \in \mathcal{K}_{\log}} E_k, \quad \mathcal{K}_E^* := \arg \min_{k \in \mathcal{K}_{\log}} E_k.$$

Since u^* is unknown, E_* and $\mathcal{K}_E^*$ are unavailable in practical computations.

At auxiliary level m , define

$$\mathcal{K}_m^* := \arg \min_{k \in \mathcal{K}_{\log}} \eta_{m,k}.$$

A reference-free checkpoint may therefore be selected as any $k_m^* \in \mathcal{K}_m^*$. This rule uses only the unconditional lower monitor and is independent of saturation, which enters only the finite-resolution guarantees developed later.

Lemma 2 (Uniform recovery over the logged set). *Under the assumptions of Corollary 2, define*

$$\varepsilon_m := \max_{k \in \mathcal{K}_{\log}} (E_k - \eta_{m,k}).$$

Then

$$\varepsilon_m = \max_{k \in \mathcal{K}_{\log}} |E_k - \eta_{m,k}| \longrightarrow 0 \quad \text{as } m \rightarrow \infty.$$

Proof. For every fixed $k \in \mathcal{K}_{\log}$, Corollary 2 gives

$$0 \leq E_k - \eta_{m,k} \longrightarrow 0.$$

Since $\mathcal{K}_{\log}$ is finite, the maximum of these finitely many convergent defects also tends to zero. $\square$

Thus,

$$0 \leq E_k - \eta_{m,k} \leq \varepsilon_m \quad \forall k \in \mathcal{K}_{\log},$$

with $\varepsilon_m \rightarrow 0$. Combined with monitor minimality, this uniform control yields value consistency.

Proposition 9 (Value consistency of monitor-based selection). *Let $k_m^* \in \mathcal{K}_m^*$ be any monitor-minimizing checkpoint. Then*

$$0 \leq E_{k_m^*} - E_* \leq \varepsilon_m.$$

Consequently,

$$E_{k_m^*} \longrightarrow E_* \quad \text{as } m \rightarrow \infty.$$

Proof. Let $k_E^* \in \mathcal{K}_E^*$. By monitor minimality, the lower-bound property, and the definition of ε_m ,

$$E_{k_m^*} \leq \eta_{m,k_m^*} + \varepsilon_m \leq \eta_{m,k_E^*} + \varepsilon_m \leq E_* + \varepsilon_m.$$

Since $E_* \leq E_{k_m^*}$, the stated bound follows. $\square$

Thus, monitor minimization is asymptotically consistent in value. Finiteness of the archive strengthens this to eventual logged-oracle selection.

Corollary 3 (Eventual logged-oracle selection). *Under the assumptions of Corollary 2, there exists m_0 such that*

$$\mathcal{K}_m^* \subseteq \mathcal{K}_E^* \quad \forall m \geq m_0.$$

If the logged energy oracle is unique,

$$\mathcal{K}_E^* = \{k_E^*\},$$

then

$$\mathcal{K}_m^* = \{k_E^*\} \quad \forall m \geq m_0.$$

Proof. If $\mathcal{K}_E^* = \mathcal{K}_{\log}$, the conclusion is immediate. Otherwise, finiteness of the archive gives the positive gap

$$\delta_E := \min_{k \in \mathcal{K}_{\log} \setminus \mathcal{K}_E^*} (E_k - E_*) > 0.$$

By Lemma 2, there exists m_0 such that $\varepsilon_m < \delta_E$ for every $m \geq m_0$. For any $k_m^* \in \mathcal{K}_m^*$, Proposition 9 gives

$$E_{k_m^*} - E_* \leq \varepsilon_m < \delta_E.$$

No checkpoint outside $\mathcal{K}_E^*$ can satisfy this inequality. Hence $k_m^* \in \mathcal{K}_E^*$. The unique-oracle case follows immediately. $\square$

Remark 6 (Scope of logged-oracle recovery). Proposition 9 and Corollary 3 concern the oracle over the prescribed archive $\mathcal{K}_{\log}$. If this archive contains a minimizer of the energy error over the computed training trajectory, then its logged oracle is also trajectory-wide. Whether such a minimizer is retained is the distinct archive-coverage question addressed in Section 5.

Remark 7 (Geometric removal of the ordering obstruction). If $\mathcal{K}_E^* = \mathcal{K}_{\log}$, there is no oracle–non-oracle ordering to resolve. Otherwise, define

$$\gamma_E := \min_{k \in \mathcal{K}_{\log} \setminus \mathcal{K}_E^*} (E_k^2 - E_*^2) > 0.$$

For any $s \in \mathcal{K}_E^*$ and $k \in \mathcal{K}_{\log} \setminus \mathcal{K}_E^*$, the orthogonal decomposition gives

$$\eta_{m,k}^2 - \eta_{m,s}^2 \geq \gamma_E - \max_{j \in \mathcal{K}_{\log}} \rho_{m,j}^2.$$

Since the archive is finite and $\rho_{m,j} \rightarrow 0$ for every fixed j ,

$$\max_{j \in \mathcal{K}_{\log}} \rho_{m,j}^2 \rightarrow 0.$$

Hence, for all sufficiently large m ,

$$\eta_{m,s} < \eta_{m,k} \quad \forall s \in \mathcal{K}_E^*, \quad \forall k \in \mathcal{K}_{\log} \setminus \mathcal{K}_E^*.$$

Thus, conforming refinement makes the unresolved projection defects uniformly too small to offset the oracle–non-oracle energy gap, eliminating the finite-resolution rank-reversal mechanism.

The preceding results provide an asymptotic consistency statement: under common conforming refinement of the finite archive, monitor minimization eventually selects only logged energy-oracle checkpoints. The proof of Corollary 3 makes the mechanism explicit. Whenever

$$\varepsilon_m < \delta_E, \quad \delta_E := \min_{k \in \mathcal{K}_{\log} \setminus \mathcal{K}_E^*} (E_k - E_*),$$

one has

$$\mathcal{K}_m^* \subseteq \mathcal{K}_E^*.$$

This asymptotic guarantee does not provide a directly verifiable finite-resolution criterion. Both ε_m and δ_E depend on the inaccessible exact energy errors E_k , and no computable convergence rate for ε_m is assumed. Hence, observing the same monitor minimizer over finitely many auxiliary levels does not imply that $\varepsilon_m < \delta_E$; the ordering obstruction may disappear only under further refinement.

Such stabilization nevertheless provides an empirical indication of selection robustness under the tested auxiliary refinements. In experiments with reference information, the ratios $\eta_{m,k}/E_k$ quantify how closely the monitors recover the corresponding energy errors. Without reference information, however, stabilization alone is not a certificate, motivating the computable finite-resolution guarantees derived next from the hierarchical lower–upper intervals.

4.3 Finite-resolution near-oracle control and logged-oracle certification

Throughout this subsection, the finite archive $\mathcal{K}_{\log}$ is prescribed before selection and remains fixed under auxiliary refinement. Changing the archive changes the target oracle and therefore defines a distinct selection problem. Write

$$E_{\log}^* := \min_{k \in \mathcal{K}_{\log}} E_k.$$

Fix a nested auxiliary pair

$$V_m \subset V_M$$

and a prescribed factor $q \in (0, 1)$. The lower-monitor selection rule is independent of q , which enters only through the conditional upper estimates. For each $k \in \mathcal{K}_{\log}$, define

$$\mathcal{U}_{m,M,q,k} := \mathcal{U}_{m,M,q}(u_k).$$

Whenever saturation holds at checkpoint k ,

$$\eta_{m,k} \leq E_k \leq \mathcal{U}_{m,M,q,k}.$$

Proposition 10 (Finite-resolution near-oracle control). *Let*

$$k_m^* \in \mathcal{K}_m^*$$

be selected by the lower monitor. Assume that saturation holds at k_m^ and that*

$$\eta_{m,k_m^*} > 0.$$

Then

$$\frac{E_{k_m^*}}{E_{\log}^*} \leq \frac{\mathcal{U}_{m,M,q,k_m^*}}{\eta_{m,k_m^*}}.$$

Proof. Let $k_E^* \in \mathcal{K}_E^*$. By monitor minimality and the lower-bound property,

$$\eta_{m,k_m^*} \leq \eta_{m,k_E^*} \leq E_{\log}^*.$$

Since $\eta_{m,k_m^*} > 0$, one has $E_{\log}^* > 0$. Saturation at k_m^* gives

$$E_{k_m^*} \leq \mathcal{U}_{m,M,q,k_m^*}.$$

Dividing the two bounds yields the result. $\square$

Corollary 4 (Exact logged-oracle certification by interval separation). *Let $s \in \mathcal{K}_{\log}$, and assume that saturation holds at s . If*

$$\mathcal{U}_{m,M,q,s} < \min_{\substack{k \in \mathcal{K}_{\log} \\ k \neq s}} \eta_{m,k},$$

then s is the unique logged energy-oracle checkpoint:

$$E_s < E_k \quad \forall k \in \mathcal{K}_{\log}, \quad k \neq s.$$

Proof. For every $k \in \mathcal{K}_{\log} \setminus \{s\}$,

$$E_s \leq \mathcal{U}_{m,M,q,s} < \eta_{m,k} \leq E_k.$$

$\square$

Operational selection procedure At a given auxiliary resolution, evaluate $\eta_{m,k}$ over the prescribed archive and select

$$k_m^* \in \arg \min_{k \in \mathcal{K}_{\log}} \eta_{m,k}.$$

The hierarchical upper estimate $\mathcal{U}_{m,M,q,k_m^*}$ is then computed at the selected checkpoint. Under saturation at k_m^* , the separation condition

$$\mathcal{U}_{m,M,q,k_m^*} < \min_{\substack{k \in \mathcal{K}_{\log} \\ k \neq k_m^*}} \eta_{m,k}$$

certifies k_m^* as the unique logged energy oracle.

If separation is not attained and $\eta_{m,k_m^*} > 0$, the same saturation hypothesis yields the computable factor

$$\Gamma_{\log,k_m^*} := \frac{\mathcal{U}_{m,M,q,k_m^*}}{\eta_{m,k_m^*}},$$

with

$$\frac{E_{k_m^*}}{E_{\log}^*} \leq \Gamma_{\log,k_m^*}.$$

If this bound is sufficiently sharp for the intended use, k_m^* may be retained; otherwise, the lower auxiliary space is refined, the monitors are reevaluated over the same archive, and the procedure is repeated with a further nested enrichment.

Scope and contribution In the present setting, selection is restricted to candidates generated by the optimization run and retained in the prescribed archive; it cannot compensate for a trajectory that fails to contain a sufficiently accurate approximation. The symmetric coercive Ritz structure nevertheless yields an important alignment:

$$J(u_k) - J(u^*) = \frac{1}{2} E_k^2.$$

Hence the exact Ritz functional and the energy error induce the same ordering over admissible candidates. Practical empirical or quadrature-based evaluations may, however, perturb this ordering unless the associated approximation errors are sufficiently controlled; see the discussion following Proposition 2.

The contribution is therefore to recover, without a reference solution, the minimum-energy-error candidate contained in the prescribed archive. Conforming refinement yields eventual logged-oracle recovery, while the hierarchical bounds provide finite-resolution near-oracle control and, under saturation, exact certification by interval separation. Whether the archive omits a better iterate from a denser computed trajectory is the distinct coverage question addressed next.

5 Logging resolution and trajectory-wide guarantees

A logging schedule determines the finite archive over which checkpoint selection is performed. Consequently, the corresponding logged oracle depends on the logging resolution: a coarser archive may exclude lower-energy iterates and thereby increase the smallest attainable energy error among the retained checkpoints.

For a uniform logging stride h dividing T , let

$$\mathcal{K}_h := \{0, h, 2h, \dots, T\}, \quad E_h^* := \min_{k \in \mathcal{K}_h} E_k.$$

If h_2 is a multiple of h_1 , then

$$\mathcal{K}_{h_2} \subseteq \mathcal{K}_{h_1}, \quad E_{h_1}^* \leq E_{h_2}^*.$$

Thus, coarser logging cannot improve the best energy-error level available within the archive.

To quantify this coverage effect, let $\mathcal{K}_{\text{traj}}$ be a prescribed finite comparison trajectory, which may in particular consist of a denser set of iterates from the same optimization run, and fix an archive

$$\mathcal{K}_{\log} \subseteq \mathcal{K}_{\text{traj}}.$$

Define

$$E_{\log}^* := \min_{k \in \mathcal{K}_{\log}} E_k, \quad E_{\text{traj}}^* := \min_{j \in \mathcal{K}_{\text{traj}}} E_j.$$

Then

$$0 \leq E_{\log}^* - E_{\text{traj}}^*,$$

and this difference measures the loss induced by archive subsampling relative to the comparison trajectory.

5.1 Competitive archive coverage and trajectory-wide control

Assume that $\eta_{m,j}$ is available for every $j \in \mathcal{K}_{\text{traj}}$, and select

$$s \in \arg \min_{k \in \mathcal{K}_{\log}} \eta_{m,k}.$$

Fix a nested pair $V_m \subset V_M$ and a prescribed $q \in (0, 1)$, and assume that saturation holds at s . Define

$$U_s := \mathcal{U}_{m,M,q,s}, \quad E_s \leq U_s.$$

The available lower–upper information allows trajectory iterates that cannot improve upon s to be excluded. Define

$$\mathcal{K}_{\text{comp},s} := \{j \in \mathcal{K}_{\text{traj}} : \eta_{m,j} \leq U_s\}. \quad (26)$$

Indeed, if $j \notin \mathcal{K}_{\text{comp},s}$, then

$$E_j \geq \eta_{m,j} > U_s \geq E_s,$$

so j cannot improve upon the selected checkpoint. Conversely, if $j^* \in \arg \min_{j \in \mathcal{K}_{\text{traj}}} E_j$ is a trajectory oracle, then

$$\eta_{m,j^*} \leq E_{\text{traj}}^* \leq E_s \leq U_s.$$

Hence every trajectory oracle belongs to $\mathcal{K}_{\text{comp},s}$, and trajectory-wide coverage may be restricted to this competitive set.

Define the competitive coverage radius by

$$r_{\text{comp},s} := \max_{j \in \mathcal{K}_{\text{comp},s}} \min_{k \in \mathcal{K}_{\log}} \|u_j - u_k\|_a. \quad (27)$$

It measures the largest energy distance from a competitive trajectory iterate to its nearest retained checkpoint. These pairwise distances do not require access to u^* .

Proposition 11 (Trajectory-wide control through competitive coverage). *Under the preceding assumptions,*

$$0 \leq E_{\log}^* - E_{\text{traj}}^* \leq r_{\text{comp},s}. \quad (28)$$

Moreover,

$$0 \leq E_s - E_{\text{traj}}^* \leq \underbrace{U_s - \eta_{m,s}}_{\text{auxiliary-resolution gap}} + \underbrace{r_{\text{comp},s}}_{\text{archive-coverage loss}}. \quad (29)$$

The trajectory-oracle level satisfies

$$E_{\text{traj}}^* \geq \eta_{m,s} - r_{\text{comp},s}. \quad (30)$$

Hence, if

$$r_{\text{comp},s} < \eta_{m,s},$$

then

$$\Gamma_{\text{traj},s} := \frac{U_s}{\eta_{m,s} - r_{\text{comp},s}} \quad (31)$$

is well defined and satisfies

$$\frac{E_s}{E_{\text{traj}}^*} \leq \Gamma_{\text{traj},s}. \quad (32)$$

If, in addition,

$$U_s < \min_{k \in \mathcal{K}_{\log} \setminus \{s\}} \eta_{m,k}, \quad (33)$$

then s is the unique logged energy oracle and

$$E_s = E_{\log}^*, \quad 0 \leq E_s - E_{\text{traj}}^* \leq r_{\text{comp},s}. \quad (34)$$

Proof. Let $j^* \in \mathcal{K}_{\text{traj}}$ attain E_{traj}^* . Since $j^* \in \mathcal{K}_{\text{comp},s}$, choose $k^* \in \mathcal{K}_{\log}$ such that

$$\|u_{j^*} - u_{k^*}\|_a \leq r_{\text{comp},s}.$$

The triangle inequality gives

$$E_{k^*} \leq E_{\text{traj}}^* + r_{\text{comp},s}.$$

Since $E_{\text{traj}}^* \leq E_{\log}^* \leq E_{k^*}$, (28) follows.

Let $k_{\log}^*$ attain $E_{\log}^*$. Monitor minimality and the lower-bound property give

$$\eta_{m,s} \leq \eta_{m,k_{\log}^*} \leq E_{\log}^*.$$

Together with (28), this yields

$$E_{\text{traj}}^* \geq \eta_{m,s} - r_{\text{comp},s},$$

proving (30). Since $E_s \leq U_s$,

$$E_s - E_{\text{traj}}^* \leq U_s - \eta_{m,s} + r_{\text{comp},s},$$

which gives (29). If $r_{\text{comp},s} < \eta_{m,s}$, division by the positive lower bound in (30) yields (32).

Finally, (33) and Corollary 4 certify s as the unique logged energy oracle. Hence $E_s = E_{\log}^*$, and (34) follows from (28). $\square$

The trajectory-wide suboptimality admits the exact decomposition

$$E_s - E_{\text{traj}}^* = (E_s - E_{\log}^*) + (E_{\log}^* - E_{\text{traj}}^*).$$

The first term is the loss incurred by monitor-based selection within the prescribed archive, whereas the second is the loss caused by archive subsampling relative to the comparison trajectory. Proposition 11 bounds these two contributions by

$$0 \leq E_s - E_{\log}^* \leq U_s - \eta_{m,s}, \quad 0 \leq E_{\log}^* - E_{\text{traj}}^* \leq r_{\text{comp},s}.$$

Hence (29) separates the trajectory-wide suboptimality into an auxiliary-resolution contribution and an archive-coverage contribution.

5.2 Competitive coverage evaluation and path-length bounds

When the required pairwise energy distances are available, the competitive coverage radius can be evaluated directly. Define

$$d_a(j, \mathcal{K}_{\log}) := \min_{k \in \mathcal{K}_{\log}} \|u_j - u_k\|_a.$$

Then

$$r_{\text{comp},s} = \max_{j \in \mathcal{K}_{\text{comp},s}} d_a(j, \mathcal{K}_{\log}).$$

Direct evaluation may, however, require many pairwise energy-distance computations when the comparison trajectory or retained archive is large. Successive trajectory increments provide an alternative based only on local inter-iterate distances or their upper bounds, for example when these are available from densely recorded consecutive checkpoints or from bounds on successive iterate displacements.

Suppose that

$$\mathcal{K}_{\text{traj}} = \{0, \dots, T\}, \quad \mathcal{K}_{\log} = \{t_0, \dots, t_N\}, \quad 0 = t_0 < \dots < t_N = T,$$

and that

$$\|u_{j+1} - u_j\|_a \leq \bar{d}_j, \quad j = 0, \dots, T-1.$$

Corollary 5 (Competitive path-length bound). *Under the preceding assumptions,*

$$r_{\text{comp},s} \leq \bar{r}_{\text{comp},s}, \tag{35}$$

where

$$\bar{r}_{\text{comp},s} := \max_{\substack{0 \leq i < N \\ j \in \mathcal{K}_{\text{comp},s} \cap \{t_i, \dots, t_{i+1}\}}} \min \left\{ \sum_{\ell=t_i}^{j-1} \bar{d}_\ell, \sum_{\ell=j}^{t_{i+1}-1} \bar{d}_\ell \right\}, \tag{36}$$

with empty sums interpreted as zero. For a uniform stride h and $\bar{d}_j \leq d_{\max}$,

$$r_{\text{comp},s} \leq \left\lfloor \frac{h}{2} \right\rfloor d_{\max}. \tag{37}$$

Proof. For $j \in \{t_i, \dots, t_{i+1}\}$, the triangle inequality gives

$$\|u_j - u_{t_i}\|_a \leq \sum_{\ell=t_i}^{j-1} \bar{d}_\ell, \quad \|u_j - u_{t_{i+1}}\|_a \leq \sum_{\ell=j}^{t_{i+1}-1} \bar{d}_\ell.$$

Since both endpoints are retained, the distance from u_j to the archive is bounded by the smaller path length. Maximizing over the competitive set gives (35). For uniform stride h , every index is at most $\lfloor h/2 \rfloor$ increments from a retained endpoint, yielding (37). $\square$

The path-length construction yields the valid upper bound

$$r_{\text{comp},s} \leq \bar{r}_{\text{comp},s},$$

and, for uniform logging, makes the dependence on the stride h explicit through (37). Direct evaluation of $r_{\text{comp},s}$, when available, avoids this additional relaxation, whereas the path-length bound may therefore be less sharp. More generally, any valid upper bound $\hat{r}_{\text{comp},s} \geq r_{\text{comp},s}$ may replace $r_{\text{comp},s}$ in the trajectory-wide guarantees, yielding correspondingly conservative bounds.

Thus, evaluating or bounding $r_{\text{comp},s}$ quantifies the price of archive subsampling. This enables the logging stride to be chosen by balancing sparsity against trajectory-wide accuracy.

5.3 Logging-resolution design with trajectory-wide control

Coarser logging reduces the number of retained checkpoints and can make the interval-separation condition (33) easier to satisfy by reducing the competitor set; however, it may also omit lower-energy trajectory iterates. The objective is therefore to determine how far the logging stride can be increased while keeping the resulting trajectory-wide loss under quantitative control.

Let $\mathcal{H}$ be a finite family of candidate uniform strides, chosen so that the corresponding archives $(\mathcal{K}_h)_{h \in \mathcal{H}}$ are nested, with larger h corresponding to sparser logging. All candidates are assessed relative to the same prescribed comparison trajectory $\mathcal{K}_{\text{traj}}$.

For each $h \in \mathcal{H}$, select

$$s_h \in \arg \min_{k \in \mathcal{K}_h} \eta_{m,k},$$

and, assuming saturation at s_h , define

$$U_h := \mathcal{U}_{m,M,q,s_h}, \quad a_h := U_h - \eta_{m,s_h}, \quad r_h := r_{\text{comp},s_h},$$

where r_h is evaluated relative to $\mathcal{K}_{\log} = \mathcal{K}_h$. Proposition 11 gives

$$0 \leq E_h^* - E_{\text{traj}}^* \leq r_h, \quad 0 \leq E_{s_h} - E_{\text{traj}}^* \leq a_h + r_h.$$

Thus, r_h bounds the loss induced by archive coarsening, whereas $a_h + r_h$ bounds the trajectory-wide suboptimality of the checkpoint selected from that archive. If only a valid upper bound $\hat{r}_h \geq r_h$ is available, it may replace r_h in these guarantees conservatively.

Whenever $r_h < \eta_{m,s_h}$, the corresponding relative guarantee is

$$\frac{E_{s_h}}{E_{\text{traj}}^*} \leq \Gamma_{\text{traj},s_h} := \frac{U_h}{\eta_{m,s_h} - r_h}.$$

For prescribed tolerances $\tau_{\log} > 0$ and $\tau_{\text{tot}} > 0$, the preceding bounds yield

$$h_{\text{cov}}^* := \max \{h \in \mathcal{H} : r_h \leq \tau_{\log}\}, \quad (38)$$

and

$$h_{\text{sel}}^* := \max \{h \in \mathcal{H} : a_h + r_h \leq \tau_{\text{tot}}\}. \quad (39)$$

Whenever the corresponding feasible sets are nonempty, these are the sparsest tested resolutions satisfying, respectively,

$$E_{h_{\text{cov}}^*}^* \leq E_{\text{traj}}^* + \tau_{\log}, \quad E_{s_{h_{\text{sel}}^*}} \leq E_{\text{traj}}^* + \tau_{\text{tot}}.$$

A relative alternative is to retain the largest tested h satisfying

$$\Gamma_{\text{traj},s_h} \leq 1 + \varepsilon,$$

which guarantees

$$E_{s_h} \leq (1 + \varepsilon) E_{\text{traj}}^*.$$

Candidate strides can be assessed by deterministic subsampling of a single densely recorded comparison run, without retraining or modifying the optimization schedule. The quantities r_h , $a_h + r_h$, and Γ_{traj,s_h} quantify how far logging can be coarsened: without a prescribed accuracy criterion they expose the sparsity-accuracy trade-off, whereas with one they identify the sparsest tested resolution satisfying it.

5.4 Exact oracle inclusion and trajectory-oracle certification

The preceding analysis quantifies the accuracy loss induced by archive coarsening and thereby provides a basis for choosing the logging resolution. A stronger question is whether a coarsened archive has incurred any loss at all, that is, whether it still contains a trajectory oracle. We therefore derive exact certificates for trajectory-oracle inclusion and, more strongly, for the selected checkpoint itself to be a trajectory oracle. When $\mathcal{K}_{\text{traj}}$ contains the full recorded optimization trajectory, these certificates respectively show that sparse

logging has preserved, or that the selected checkpoint attains, the best energy-error level reached along that trajectory.

To compare the best retained and omitted trajectory iterates, define

$$L_h^{\text{out}} := \min_{j \in \mathcal{K}_{\text{traj}} \setminus \mathcal{K}_h} \eta_{m,j},$$

with the convention $L_h^{\text{out}} = +\infty$ if $\mathcal{K}_h = \mathcal{K}_{\text{traj}}$. When valid hierarchical upper estimates $U_k := \mathcal{U}_{m,M,q,k}$ are available for all $k \in \mathcal{K}_h$, define

$$B_h := \min_{k \in \mathcal{K}_h} U_k.$$

Then L_h^{out} is a lower bound on the best energy error among omitted trajectory iterates, whereas B_h is an upper bound on the best energy error attained within the archive.

Proposition 12 (Exact trajectory-oracle certificates). *For a candidate archive $\mathcal{K}_h$, the following statements hold.*

1. *If valid upper estimates U_k are available for all $k \in \mathcal{K}_h$ and*

$$B_h < L_h^{\text{out}}, \tag{40}$$

then $\mathcal{K}_h$ contains a trajectory oracle, and hence

$$E_h^* = E_{\text{traj}}^*. \tag{41}$$

2. *If U_h is a valid upper estimate at s_h and*

$$U_h < \min_{j \in \mathcal{K}_{\text{traj}} \setminus \{s_h\}} \eta_{m,j}, \tag{42}$$

then s_h is the unique trajectory oracle, so that

$$E_{s_h} = E_h^* = E_{\text{traj}}^*. \tag{43}$$

Proof. Choose $\bar{k} \in \mathcal{K}_h$ such that $U_{\bar{k}} = B_h$. Under (40), every $j \in \mathcal{K}_{\text{traj}} \setminus \mathcal{K}_h$ satisfies

$$E_{\bar{k}} \leq U_{\bar{k}} = B_h < L_h^{\text{out}} \leq \eta_{m,j} \leq E_j.$$

Thus no omitted checkpoint can attain the trajectory minimum, proving (41).

Under (42),

$$E_{s_h} \leq U_h < \eta_{m,j} \leq E_j \quad \forall j \in \mathcal{K}_{\text{traj}} \setminus \{s_h\}.$$

Hence s_h is the unique trajectory oracle, which yields (43). $\square$

For nested candidate archives assessed with the same valid checkpointwise upper estimates, B_h cannot increase under archive refinement, whereas L_h^{out} cannot decrease. Hence (40), once satisfied, remains valid under further refinement. The sparsest tested archive certified to contain a trajectory oracle is therefore

$$h_{\text{incl}}^* := \max \{h \in \mathcal{H} : B_h < L_h^{\text{out}}\}, \tag{44}$$

whenever the feasible set is nonempty.

If a valid upper estimate is available only at the selected checkpoint, the selected-only condition

$$U_h < L_h^{\text{out}}$$

still certifies

$$E_h^* = E_{\text{traj}}^*.$$

Likewise, the sparsest tested archive for which the selected checkpoint is certified as the unique trajectory oracle is

$$h_{\text{cert}}^* := \max \left\{ h \in \mathcal{H} : U_h < \min_{j \in \mathcal{K}_{\text{traj}} \setminus \{s_h\}} \eta_{m,j} \right\}, \tag{45}$$

whenever the feasible set is nonempty.

Together, these criteria yield a hierarchy from quantitative control to exact certification: $r_h \leq \tau_{\log}$ controls the loss caused by archive coarsening, $a_h + r_h \leq \tau_{\text{tot}}$ controls the trajectory-wide suboptimality of the selected checkpoint, $B_h < L_h^{\text{out}}$ certifies preservation of a trajectory oracle, and (42) certifies the selected checkpoint itself as the unique trajectory oracle.

5.5 Joint logging and auxiliary refinement

The preceding results provide finite-resolution guarantees and exact certificates. We now consider the complementary asymptotic regime in which logging and auxiliary resolutions are refined simultaneously.

Corollary 6 (Trajectory-oracle recovery under joint refinement). *Consider a sequence of logging and auxiliary configurations, all assessed relative to the same finite comparison trajectory. Let s_n be the selected checkpoint, let U_n be a valid upper estimate at s_n , and define*

$$a_n := U_n - \eta_{m_n, s_n}, \quad r_n := r_{\text{comp}, s_n}.$$

If

$$a_n \longrightarrow 0, \quad r_n \longrightarrow 0,$$

then

$$E_{s_n} \longrightarrow E_{\text{traj}}^*.$$

If $E_{\text{traj}}^* > 0$, then $\Gamma_{\text{traj}, s_n}$ is well defined for all sufficiently large n and

$$\Gamma_{\text{traj}, s_n} \longrightarrow 1.$$

Proof. Proposition 11 gives

$$0 \leq E_{s_n} - E_{\text{traj}}^* \leq a_n + r_n,$$

so $E_{s_n} \rightarrow E_{\text{traj}}^*$. Moreover,

$$\eta_{m_n, s_n} \leq E_{s_n} \leq U_n, \quad U_n - \eta_{m_n, s_n} = a_n,$$

hence

$$\eta_{m_n, s_n} \longrightarrow E_{\text{traj}}^*, \quad U_n \longrightarrow E_{\text{traj}}^*.$$

If $E_{\text{traj}}^* > 0$, then

$$\eta_{m_n, s_n} - r_n \longrightarrow E_{\text{traj}}^* > 0,$$

and therefore

$$\Gamma_{\text{traj}, s_n} = \frac{U_n}{\eta_{m_n, s_n} - r_n} \longrightarrow 1.$$

□

Corollary 6 yields trajectory-oracle consistency for the complete selection procedure relative to the prescribed comparison trajectory. The auxiliary-resolution gap a_n controls selection within the logged archive, while r_n controls the loss due to archive coverage; their joint decay is sufficient for the selected checkpoint to recover the trajectory-oracle level asymptotically.

6 Numerical validation

This section evaluates the conforming Riesz monitor η_m as a post-training error-assessment and checkpoint-selection criterion. The experiments examine energy-scale calibration, reference-free oracle selection, auxiliary refinement, logging resolution and trajectory coverage, and applicability to a non-manufactured benchmark. Reference solutions are used only for validation and enter neither training, Riesz reconstruction, checkpoint selection, nor certification criteria.

We denote by

$$E_{\text{ref}} = \begin{cases} E_{\text{quad}} := \|u_\theta - u^*\|_{a, \text{quad}}, & \text{manufactured benchmarks,} \\ E_{\text{FEM}} := \|u_\theta - u_{\text{FEM}}^{\text{ref}}\|_a, & \text{perforated plate,} \end{cases}$$

the numerical validation energy error. Thus η_m/E_{ref} measures monitor calibration. For manufactured problems it approximates the fraction of the exact energy error resolved by the conforming reconstruction; for the perforated plate it provides an external calibration ratio against the independent refined FEM reference.

Table 1: Experimental protocols. All networks use hyperbolic tangent activations and Adam with learning rate 10^{-3} ; each run uses a fixed independent validation set of 2×10^4 points for J_{val} , and each benchmark is repeated over five seeds $0, \dots, 4$. The notation 64^4 denotes four hidden layers of width 64, and M_{op} denotes the operational auxiliary level.

Family	Network	Steps	Log	Training points	Validation reference	Auxiliary hierarchy
Scalar diffusion	$2-64^4-1$	2000	50	10^4	E_{quad} : 96^2 , 3×3 Gauss	6, 12, 24, 48, $M_{\text{op}} = 24$
Elasticity	$2-80^4-2$	3000	100	12000	E_{quad} : 96^2 , 3×3 Gauss	6, 12, 24, 48, $M_{\text{op}} = 24$
L-shaped domain	$2-80^4-1$	4000	100	15000	E_{quad} : 128^2 , 3×3 Gauss	8, 16, 32, 64, 128, $M_{\text{op}} = 64$
Perforated plate	$2-80^4-2$	4000	100	15000	FEM 256 (check: 192)	24, 48, 96, $M_{\text{op}} = 48$

We use the hierarchical gap $\delta_{m,M}$ and conditional upper estimate $\mathcal{U}_{m,M,q}$ from Subsection 3.3, with $q = 0.9$ unless stated otherwise. Monitor-based selection is independent of q ; sensitivity of the conditional certificates and upper estimates to q is reported in A. For validation and refinement diagnostics we also use

$$q_{\text{obs}} := \left(\frac{E_{\text{quad}}^2 - \eta_M^2}{E_{\text{quad}}^2 - \eta_m^2} \right)^{1/2}, \quad \gamma_{m,M,L} := \frac{\delta_{M,L}}{\delta_{m,M}}, \quad D_c := \frac{E_{\text{ref},k_c}}{E_{\text{ref},k_{\text{oracle}}}},$$

whenever defined. Here $q_{\text{obs}} \leq q$ checks saturation against the manufactured reference at the tested checkpoint, while $\gamma_{m,M,L} < 1$ indicates decay of successive newly resolved Riesz components but does not itself verify saturation. The factor D_c measures the energy deterioration of criterion c relative to the oracle of the same logged archive; $D_c = 1$ denotes oracle-level selection with respect to the validation reference. When E_{quad} is available, we also report $\rho_{m,\text{rel}}^{\text{quad}} := (E_{\text{quad}}^2 - \eta_m^2)^{1/2} / E_{\text{quad}}$ as an external measure of the unresolved Riesz component. The experimental protocols are summarized in Table 1.

Unless stated otherwise, reported values of the form $x \pm y$ denote the mean and one sample standard deviation over the five seeds.

For the manufactured square-domain tests, the auxiliary spaces are the conforming Q_1 spaces

$$V_m^{Q_1} = \{v_m \in C^0(\bar{\Omega}) \cap H_0^1(\Omega) : v_m|_K \in Q_1(K) \quad \forall K \in \mathcal{T}_m\},$$

with $[V_m^{Q_1}]^2$ in elasticity; the L-shaped benchmark uses the same construction on its active quadrilateral cells. To realize Algorithm 1 on the nested auxiliary hierarchies, we use the common finest-level Galerkin form $G_m = P_{m,L}^\top G_L P_{m,L}$ and $r_m = P_{m,L}^\top r_L$, so nestedness and monitor monotonicity hold algebraically at the assembled discrete level. The finest levels are $L = 48$ for the standard scalar and elasticity hierarchies, $L = 96$ for the additional κ_4 certification hierarchy, and $L = 128$ for the L-shaped hierarchy.

For the perforated plate, let $V_{m,\circ}^{P_1}$ denote the conforming piecewise-linear space satisfying the homogeneous essential condition on the outer boundary. We use the exactly nested vector-valued hierarchy

$$[V_{24,\circ}^{P_1}]^2 \subset [V_{48,\circ}^{P_1}]^2 \subset [V_{96,\circ}^{P_1}]^2,$$

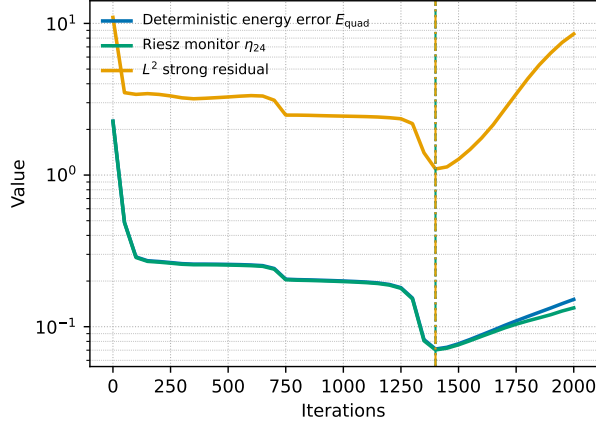
obtained by straight-sided uniform refinement of a common conforming triangular mesh fitted to the fixed polygonal hole. All levels therefore represent the same polygonal domain. The discrete-hierarchy and quadrature-order audit is reported in C. All approximations are trained with the Deep Ritz energy [20], with essential boundary conditions imposed strongly. J_{train} and J_{val} denote energy evaluations on the fixed training and independent validation sets, respectively. The strong-form residual is used only as an L^2 -type baseline. Neural training and automatic differentiation use single precision, whereas auxiliary assembly, Riesz solves, reference quadratures, and FEM computations use double precision. The operational levels M_{op} in Table 1 are those used for checkpoint ranking; enriched levels are used only for post-training refinement, calibration, and certification diagnostics.

6.1 Manufactured-reference validation benchmarks

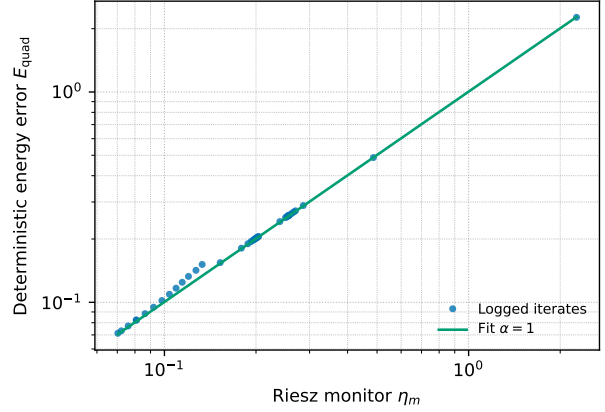
6.1.1 Scalar variable-coefficient diffusion

We first consider

$$-\nabla \cdot (\kappa \nabla u) = f \quad \text{in } \Omega = (0,1)^2, \quad u = 0 \quad \text{on } \partial\Omega,$$



(a) Training diagnostics.



(b) Calibration against E_{quad} .

Figure 1: Representative κ_1 run (seed 0): temporal tracking and calibration of the operational monitor η_{24} against E_{quad} .

Table 2: Scalar diffusion: checkpoint selection over five seeds.

Selection criterion	κ_1		κ_2		κ_3		κ_4	
	Step	D_c	Step	D_c	Step	D_c	Step	D_c
Energy oracle min E_{quad}	1390 $\pm$ 201	1.000 $\pm$ 0.000	1420 $\pm$ 175	1.000 $\pm$ 0.000	1410 $\pm$ 222	1.000 $\pm$ 0.000	1480 $\pm$ 289	1.000 $\pm$ 0.000
Riesz monitor min η_{24}	1390 $\pm$ 201	1.000 $\pm$ 0.000	1420 $\pm$ 175	1.000 $\pm$ 0.000	1410 $\pm$ 222	1.000 $\pm$ 0.000	1480 $\pm$ 289	1.000 $\pm$ 0.000
Strong residual L^2	1390 $\pm$ 201	1.000 $\pm$ 0.000	1250 $\pm$ 332	1.105 $\pm$ 0.223	1230 $\pm$ 355	1.074 $\pm$ 0.163	1300 $\pm$ 237	1.049 $\pm$ 0.109
Training Ritz loss min J_{train}	2000 $\pm$ 0	1.723 $\pm$ 0.368	2000 $\pm$ 0	1.778 $\pm$ 0.294	2000 $\pm$ 0	1.603 $\pm$ 0.389	2000 $\pm$ 0	1.529 $\pm$ 0.358
Validation Ritz loss min J_{val}	1370 $\pm$ 182	1.036 $\pm$ 0.067	1390 $\pm$ 156	1.020 $\pm$ 0.029	1420 $\pm$ 160	1.030 $\pm$ 0.040	1460 $\pm$ 268	1.012 $\pm$ 0.020

with $V_0 = H_0^1(\Omega)$ and $a_i(u, v) = \int_{\Omega} \kappa_i \nabla u \cdot \nabla v \, dx$. The first three cases use the common manufactured solution

$$u_i^*(x, y) = \sin(\pi x) \sin(\pi y), \quad i = 1, 2, 3,$$

with

$$\begin{aligned} \kappa_1 &= 1 + \frac{1}{2} \sin(2\pi x) \sin(2\pi y), & \kappa_2 &= 1 + \frac{1}{4} \sin(6\pi x) \sin(6\pi y), \\ \kappa_3 &= 1 + 0.45 \sin(10\pi x) \sin(10\pi y). \end{aligned}$$

The fourth combines the more oscillatory coefficient

$$\kappa_4 = 1 + 0.49 \sin(14\pi x) \sin(14\pi y)$$

with

$$\begin{aligned} u_4^*(x, y) &= \sin(\pi x) \sin(\pi y) + 0.35 x(1-x)y(1-y) \\ &\quad \times \exp[-80((x-0.35)^2 + (y-0.65)^2)] (1 + 0.5 \sin(2\pi x) \sin(3\pi y)). \end{aligned}$$

In each case,

$$f_i = -\nabla \cdot (\kappa_i \nabla u_i^*).$$

Figure 1 shows that η_{24} tracks the energy error on its natural scale and attains its minimum in the same late-training regime. The strong L^2 residual evolves on a different scale, as expected from Remark 2.

Across all twenty scalar runs (Table 2), η_{24} selects the same logged checkpoint as the E_{quad} -oracle. The validation Ritz loss remains close to the oracle, whereas the training loss selects the final iterate and the strong residual departs from the oracle in the three more demanding cases. This does not contradict the exact Ritz–energy identity, which applies to the continuous functional J ; the discrete evaluations J_{train} and J_{val} may perturb its ordering, as discussed after Proposition 2. None of these baselines carries the energy-dual ordering guarantee used by the Riesz criterion $k_m^* \in \arg \min_{k \in \mathcal{K}_{\log}} \eta_{m,k}$.

At the selected checkpoints (Table 3), $\eta_{24}/E_{\text{quad}} = 0.975\text{--}0.983$, increasing to $\eta_{48}/E_{\text{quad}} = 0.994\text{--}0.996$. The observed saturation factors are near 0.5, and the conditional upper estimates remain within about 5%–7% of the reference error for $q = 0.9$.

Table 3: Scalar diffusion: calibration and hierarchical diagnostics at the η_{24} -selected checkpoint.

Coefficient	Monitor calibration		Refinement / estimator diagnostics		
	$\eta_{24}/E_{\text{quad}}$	$\eta_{48}/E_{\text{quad}}$	$\gamma_{12,24,48}$	q_{obs}	$\mathcal{U}_{24,48,0.9}/E_{\text{quad}}$
κ_1	0.981 ± 0.007	0.995 ± 0.002	0.519 ± 0.004	0.503 ± 0.001	1.052 ± 0.018
κ_2	0.975 ± 0.011	0.994 ± 0.003	0.535 ± 0.019	0.506 ± 0.004	1.070 ± 0.029
κ_3	0.983 ± 0.008	0.996 ± 0.002	0.528 ± 0.013	0.506 ± 0.003	1.048 ± 0.020
κ_4	0.979 ± 0.013	0.995 ± 0.004	0.525 ± 0.028	0.509 ± 0.011	1.057 ± 0.033

Table 4: κ_4 : logged-oracle selection and interval certification for $q = 0.9$. Match uses E_{quad} only for external validation; certified counts are conditional on saturation.

Auxiliary pair	h	$ \mathcal{K}_h $	Match	Certified	$\Gamma_{\log,h}$
$V_{24} \subset V_{48}$	50	41	5/5	0/5	1.079 ± 0.048
	100	21	5/5	2/5	1.080 ± 0.048
	200	11	4/5	4/5	1.088 ± 0.065
	400	6	5/5	4/5	1.094 ± 0.066
$V_{48} \subset V_{96}$	50	41	5/5	2/5	1.021 ± 0.014
	100	21	5/5	5/5	1.021 ± 0.014
	200	11	4/5	4/5	1.025 ± 0.023
	400	6	5/5	5/5	1.027 ± 0.023

Thus, in these scalar tests, the level-24 monitor already resolves the archive ordering needed for oracle selection and incurs below-1% observed overhead (Table 10), supporting its use as the operational monitor; enrichment further sharpens energy-scale calibration.

Across all four coefficients, auxiliary refinement increases η_m/E_{quad} toward unity (Figure 3a), consistent with Corollary 2.

Having established energy-scale calibration, we next test finite-resolution certification of the selected archive minimizer.

Finite-level interval certification We use κ_4 to test finite-resolution certification across auxiliary resolutions and logging strides. The dense log is

$$\mathcal{K}_{50} = \{0, 50, \dots, 2000\},$$

and the nested subarchives are $\mathcal{K}_h = \{0, h, 2h, \dots, 2000\}$ for $h \in \{50, 100, 200, 400\}$. For the monitor-selected checkpoint s_h , let $U_h := \mathcal{U}_{m,M,q,s_h}$. We report the conditional near-oracle factor $\Gamma_{\log,h} = U_h/\eta_{m,s_h}$; interval separation certifies s_h as the unique oracle of $\mathcal{K}_h$, conditional on saturation.

Auxiliary enrichment improves certifiability (Table 4): with $V_{48} \subset V_{96}$, all five $\mathcal{K}_{100}$ selections are certified and match the observed E_{quad} -oracle. Higher certification counts are observed on several coarser subarchives, although the dependence on h is not monotone. On $\mathcal{K}_{50}$, all five selections still match but only two are certified, illustrating that failure of interval separation is inconclusive rather than evidence of incorrect selection. The single mismatch observed on $\mathcal{K}_{200}$ is not certified and has only 0.72% deterioration. No certified mismatch occurs.

Trajectory-wide archive control Certification within $\mathcal{K}_h$ does not control checkpoints omitted by logging. We therefore take $\mathcal{K}_{50}$ as the prescribed finite comparison trajectory and apply Section 5 to $h \in \{100, 200, 400\}$ using $V_{48} \subset V_{96}$ and $q = 0.9$. For external validation we report

$$\delta_{s_h}^{\text{traj}} := 100 \left(\frac{E_{\text{quad},s_h}}{\min_{j \in \mathcal{K}_{50}} E_{\text{quad},j}} - 1 \right).$$

The hatted quantities below are deterministic-quadrature realizations; they are reference-free diagnostics relative to $\mathcal{K}_{50}$, not certified continuous-energy-norm bounds unless quadrature error is controlled.

Table 5: Trajectory-wide κ_4 diagnostics relative to $\mathcal{K}_{50}$. Oracle inclusion is reported as observed/certified; in these runs the selected-trajectory-oracle counts are identical. The last column gives mean/max observed deterioration.

h	$ \mathcal{K}_h $	Oracle inclusion	$\hat{r}_h$	$\hat{a}_h + \hat{r}_h$	$\hat{\Gamma}_{\text{traj},h}$	mean/max $\delta_{s_h}^{\text{traj}}$
100	21	3/5 / 2/5	0.012 ± 0.013	0.015 ± 0.015	1.127 ± 0.111	0.22%/0.91%
200	11	1/5 / 1/5	0.032 ± 0.026	0.036 ± 0.029	1.397 ± 0.334	3.38%/7.91%
400	6	0/5 / 0/5	0.061 ± 0.041	0.065 ± 0.045	2.197 ± 1.148	8.15%/23.89%

Table 6: Elasticity: checkpoint selection over five seeds.

Selection criterion	$\mathcal{C}_1$		$\mathcal{C}_2$		$\mathcal{C}_{\text{inc}}$	
	Step	D_c	Step	D_c	Step	D_c
Energy oracle min E_{quad}	960 $\pm$ 114	1.000 $\pm$ 0.000	960 $\pm$ 55	1.000 $\pm$ 0.000	400 $\pm$ 122	1.000 $\pm$ 0.000
Riesz monitor min η_{24}	960 $\pm$ 114	1.000 $\pm$ 0.000	960 $\pm$ 55	1.000 $\pm$ 0.000	420 $\pm$ 130	1.000 $\pm$ 0.000
Strong residual L^2	680 $\pm$ 84	1.249 $\pm$ 0.141	700 $\pm$ 71	1.206 $\pm$ 0.127	320 $\pm$ 110	1.011 $\pm$ 0.016
Training Ritz loss min J_{train}	3000 $\pm$ 0	6.597 $\pm$ 2.770	3000 $\pm$ 0	6.819 $\pm$ 2.047	3000 $\pm$ 0	12.287 $\pm$ 3.022
Validation Ritz loss min J_{val}	940 $\pm$ 89	1.011 $\pm$ 0.024	940 $\pm$ 55	1.000 $\pm$ 0.001	500 $\pm$ 245	1.059 $\pm$ 0.089

The trajectory-wide results (Table 5) separate within-archive certifiability from trajectory coverage. Although $\mathcal{K}_{400}$ admits a logged-oracle certificate in all five runs with $V_{48} \subset V_{96}$, it contains no $\mathcal{K}_{50}$ -oracle. By contrast, $\mathcal{K}_{100}$ halves the archive from 41 to 21 checkpoints and limits the maximum observed trajectory deterioration to 0.91%. Coarser logging increases the reported trajectory-wide bounds and the observed deterioration. All trajectory-wide certificates obtained here satisfy the external check $q_{\text{obs}} \leq 0.509 < 0.9$. Thus auxiliary refinement controls selection within a fixed archive, whereas logging refinement controls coverage of the prescribed trajectory.

We next test the same construction in the vector-valued energy geometry of linear elasticity and under localized material contrast.

6.1.2 Plane-strain elasticity and smoothed high-contrast inclusion

We consider plane-strain elasticity on $\Omega = (0, 1)^2$, with homogeneous Dirichlet conditions, $V_0 = H_0^1(\Omega; \mathbb{R}^2)$, and

$$a(u, v) = \int_{\Omega} 2\mu(x)\varepsilon(u) : \varepsilon(v) + \lambda(x) \operatorname{div} u \operatorname{div} v \, dx,$$

where

$$\varepsilon(v) = \frac{1}{2}(\nabla v + \nabla v^{\top}), \quad \sigma(v) = 2\mu\varepsilon(v) + \lambda \operatorname{div}(v)I.$$

The manufactured displacement is

$$u^*(x, y) = \begin{pmatrix} \sin(\pi x) \sin(\pi y) \\ \frac{1}{2} \sin(2\pi x) \sin(\pi y) \end{pmatrix}, \quad -\nabla \cdot \sigma(u^*) = f.$$

With $E_0 = 1$, $\nu = 0.30$, and the corresponding baseline Lamé parameters (λ_0, μ_0) , we set $\lambda(x) = s(x)\lambda_0$, $\mu(x) = s(x)\mu_0$ and consider

$$s_1 = 1, \quad s_2 = 1 + 0.45 \sin(6\pi x) \sin(4\pi y),$$

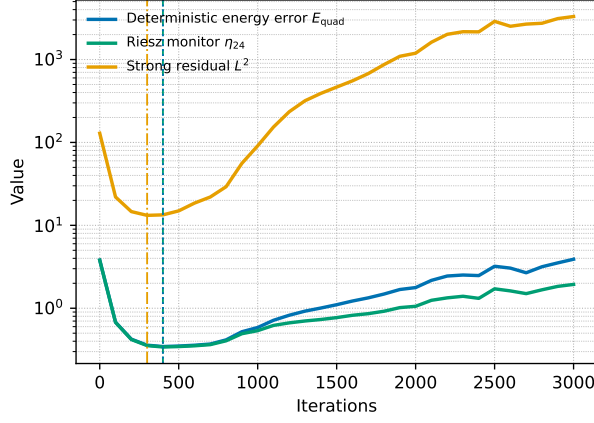
plus the smoothed high-contrast inclusion

$$s_{\text{inc}}(x, y) = 1 + 19\chi_{\varepsilon}(x, y), \quad \chi_{\varepsilon}(x, y) = \frac{1}{2} \left[1 - \tanh \left(\frac{\sqrt{(x - 0.55)^2 + (y - 0.52)^2} - 0.18}{0.035} \right) \right].$$

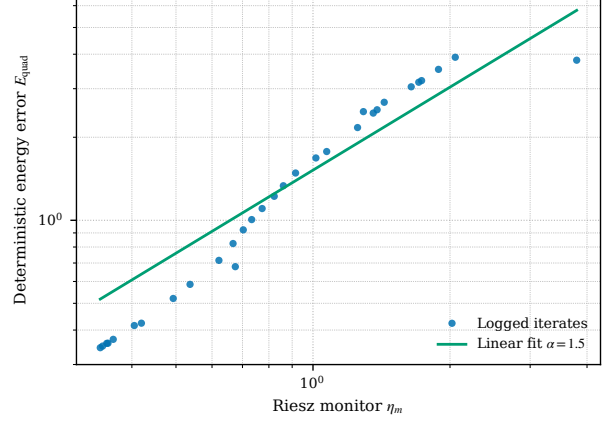
The three cases are denoted by $\mathcal{C}_1$, $\mathcal{C}_2$, and $\mathcal{C}_{\text{inc}}$.

Figure 2 shows that the vector-valued monitor remains aligned with the elastic energy error, including in the high-contrast inclusion case.

The Riesz criterion is oracle-level in all three material cases (Table 6); in one inclusion seed it selects a neighboring checkpoint with negligible deterioration ($D_c = 1.000077$). The training Ritz loss always selects



(a) Training diagnostics.



(b) Calibration against E_{quad} .

Figure 2: Representative high-contrast inclusion run: temporal tracking and calibration of the vector-valued monitor η_{24} .

Table 7: Elasticity: calibration and hierarchical diagnostics at the η_{24} -selected checkpoint.

Material	Monitor calibration		Hierarchical estimator	
	$\eta_{24}/E_{\text{quad}}$	$\eta_{48}/E_{\text{quad}}$	q_{obs}	$\mathcal{U}_{24,48,0.9}/E_{\text{quad}}$
C_1	0.971 ± 0.007	0.993 ± 0.002	0.507 ± 0.002	1.079 ± 0.019
C_2	0.973 ± 0.008	0.993 ± 0.002	0.507 ± 0.002	1.074 ± 0.020
C_{inc}	0.986 ± 0.004	0.996 ± 0.001	0.509 ± 0.002	1.040 ± 0.011

the final logged iterate with mean deterioration factors $D_c = 6.60$ – 12.29 , while independent validation is substantially closer to the oracle. As in the scalar benchmarks, the comparison with J_{val} is empirical, since finite-set evaluations may perturb the ordering of the continuous Ritz functional.

At the operational level (Table 7), the mean calibration ratio $\eta_{24}/E_{\text{quad}}$ ranges from 0.971 to 0.986; level 48 raises the mean ratio to 0.993–0.996. The external saturation check again gives $q_{\text{obs}} \simeq 0.5$, while the mean $q = 0.9$ upper-estimate ratio ranges from 1.040 to 1.079.

Across the three elasticity cases, η_m/E_{quad} again increases toward unity under enrichment (Figure 3b), in agreement with the refinement behavior established in Corollary 2.

Thus oracle-level selection and tight energy-scale calibration persist in vector-valued elasticity with localized material contrast. A complementary spatial validation in Figure 6 shows that the Riesz-projected density captures the dominant energetic concentration associated with the localized inclusion. We next stress the auxiliary-resolution requirement using a reentrant-corner singularity.

6.1.3 L-shaped-domain stress test

The final manufactured benchmark uses

$$\Omega_L = (-1, 1)^2 \setminus ([0, 1) \times (-1, 0]), \quad -\Delta u = f, \quad u = 0 \text{ on } \partial\Omega_L,$$

with

$$u^*(x, y) = (1 - x^2)(1 - y^2)r^{2/3} \sin\left(\frac{2\theta}{3}\right), \quad f = -\Delta u^*.$$

Here (r, θ) are polar coordinates centered at the reentrant corner, with $\theta \in [0, 3\pi/2]$. The reentrant-corner singularity tests checkpoint selection when uniform auxiliary spaces require substantially greater resolution.

Figure 4 shows that all monitored levels track the evolution of E_{quad} , while auxiliary enrichment improves their energy-scale calibration. The stronger resolution requirement is consistent with the reentrant-corner singularity.

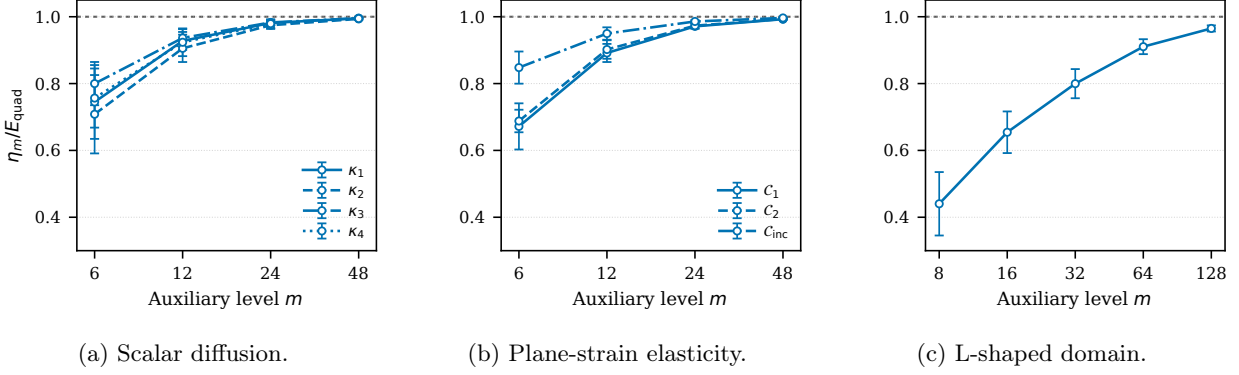


Figure 3: Auxiliary refinement at the operationally selected checkpoints for the manufactured benchmarks. Curves report the mean η_m/E_{quad} over five seeds, with error bars denoting one standard deviation. Within each run, the neural checkpoint is held fixed as m varies; the dashed line marks unit calibration.

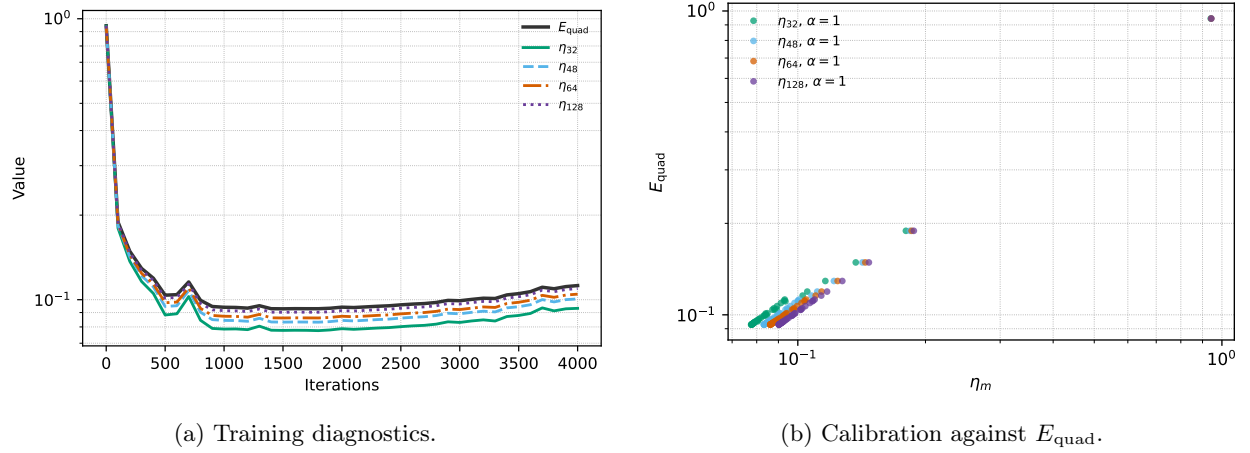


Figure 4: L-shaped-domain monitor comparison for a representative run. The conforming Riesz monitors at increasing auxiliary resolutions track the reference energy error along the logged trajectory and approach the energy-error scale under enrichment.

Table 8: L-shaped domain: checkpoint selection over five seeds. The strong residual uses the cutoff $r > 0.035$ around the reentrant corner.

Selection criterion	Selected step	D_c
min η_{32}	1900 ± 1086	1.003 ± 0.005
min η_{48}	1980 ± 1031	1.000 ± 0.001
min η_{64}	1960 ± 1031	1.000 ± 0.000
min η_{128}	1960 ± 1031	1.000 ± 0.000
Strong residual $L^2(r > 0.035)$	1640 ± 767	1.007 ± 0.006
Training Ritz loss min J_{train}	4000 ± 0	1.181 ± 0.095
Validation Ritz loss min J_{val}	1800 ± 1251	1.092 ± 0.169

The light monitor η_{32} is already near-oracle, while η_{64} and η_{128} recover the logged oracle in all five runs (Table 8). Exact logged-oracle recovery across all five runs is therefore reached only at a finer auxiliary resolution in this benchmark than in the smooth square-domain tests, consistent with Corollary 3.

At the η_{64} -selected checkpoint (Table 9), the captured fraction rises from 0.801 ± 0.044 at level 32 to 0.965 ± 0.009 at level 128, while the unresolved fraction decreases accordingly. For $V_{64} \subset V_{128}$, the external check gives $q_{\text{obs}} = 0.637 \pm 0.007 < 0.9$ and the conditional upper estimate is 1.165 ± 0.035 times E_{quad} . For the reported L-shaped runs, we use η_{64} as the operational selector and η_{128} as the enriched post-training check.

Table 9: L-shaped domain: selected auxiliary-resolution diagnostics.

Level/diagnostic	η_m/E_{quad}	$\rho_{m,\text{rel}}^{\text{quad}}$	γ or q_{obs}	$\mathcal{U}/E_{\text{quad}}$
32×32	0.801 ± 0.044	0.595 ± 0.061	—	—
64×64	0.911 ± 0.022	0.409 ± 0.050	—	—
128×128	0.965 ± 0.009	0.260 ± 0.034	—	—
(32, 64, 128)	—	—	0.729 ± 0.037	—
(64, 128)	—	—	0.637 ± 0.007	1.165 ± 0.035

Table 10: Representative cost of the operational and enriched Riesz reconstructions. Overhead is measured against the pure optimizer-update time between logged checkpoints.

Family	M_{op}	Operational call (s)	Operational overhead	Enriched level	Enriched overhead
Scalar diffusion	24	0.034–0.045	0.58–0.76%	48	12.32–15.30%
Elasticity/inclusion	24	1.881–2.045	8.01–8.72%	48	8.27–9.01%
L-shaped domain	64	0.131 ± 0.009	$0.74 \pm 0.03\%$	128	$4.04 \pm 0.74\%$

Refinement remains monotone and progressively improves energy-scale calibration (Figure 3c), consistently with Corollary 2.

Notably, $\eta_{64}/E_{\text{quad}} = 0.911 \pm 0.022$ already recovers the logged oracle in all five runs. Thus, in this benchmark, oracle recovery precedes near-unit calibration: the archive ordering is already resolved sufficiently for oracle recovery before near-exact recovery of the error scale is reached. A complementary spatial validation in Figure 7 shows that the enriched Riesz reconstruction captures the dominant energy concentration around the reentrant corner.

We next quantify the cost of the operational auxiliary levels.

6.1.4 Operational cost of checkpoint selection

The operational monitor incurs less than 1% overhead in the scalar and L-shaped tests and 8.01%–8.72% in elasticity (Table 10). These costs support its practical use for checkpoint ranking along the logged trajectory, while enriched reconstructions are reserved for tighter post-training assessment and certification.

We finally test the selector without a closed-form solution, using FEM only for external post-training validation.

6.2 Perforated plate: reference-free selection with external FEM validation

We consider plane-strain elasticity on

$$\Omega = (0, 1)^2 \setminus \overline{P_{96}(c, r)}, \quad c = (1/2, 1/2), \quad r = 0.16,$$

where $P_{96}(c, r)$ is a fixed polygonal approximation of a circular hole. The material is homogeneous with $E = 1$, $\nu = 0.30$, zero body force, and outer-boundary displacement

$$u_D(x, y) = (\varepsilon_0 x + \gamma_0 y, -\nu \varepsilon_0 y), \quad \varepsilon_0 = 5 \times 10^{-2}, \quad \gamma_0 = 2.5 \times 10^{-2},$$

while the hole boundary is traction-free. The neural approximation satisfies the prescribed outer-boundary displacement strongly, and we use the nested auxiliary hierarchy $V_{24} \subset V_{48} \subset V_{96} \subset V_0$ described above.

No closed-form solution is available. Both external FEM references are conforming vector-valued P_1 Galerkin solutions on fitted triangular meshes of the same fixed polygonal domain. The level-256 reference, used only for external post-training validation, has 179 882 displacement degrees of freedom, while the level-192 check has 109 608. The latter yields the same FEM-reference oracle in all five runs and changes checkpoint errors by at most 6.3×10^{-4} relatively. Neither FEM solve enters training, Riesz reconstruction, or checkpoint selection.

Figure 5 shows that both η_{48} and η_{96} remain closely aligned with E_{FEM} along the logged trajectory.

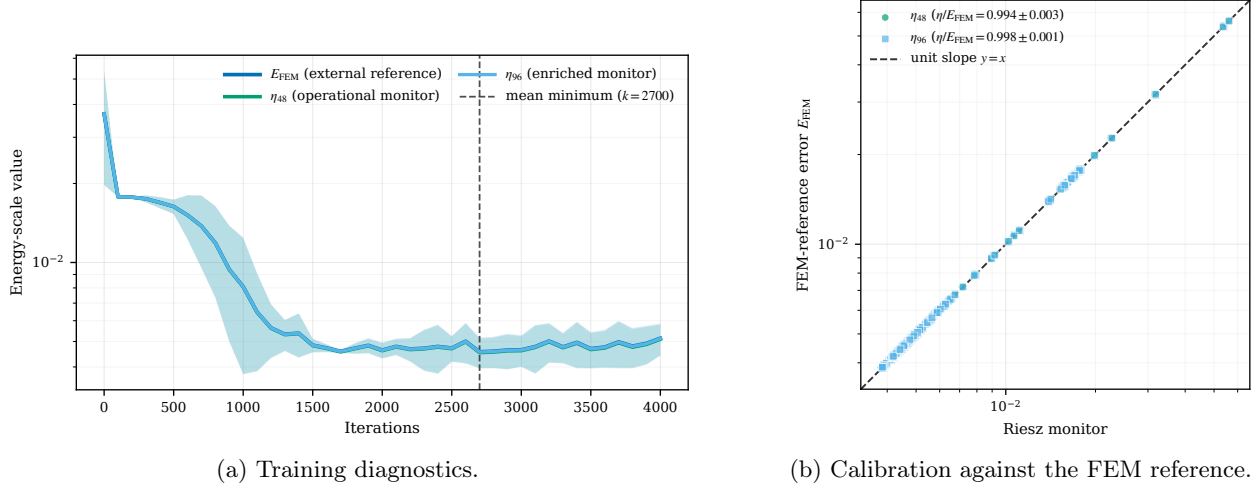


Figure 5: Perforated plate: external FEM validation of the operational η_{48} monitor over five runs.

Table 11: Perforated plate: checkpoint selection against the external FEM reference.

Criterion c	Step	D_c^{FEM}
final	4000 ± 0	1.204 ± 0.138
min E_{FEM}	2640 ± 873	1.000 ± 0.000
min η_{48}	2640 ± 873	1.000 ± 0.000
min η_{96}	2640 ± 873	1.000 ± 0.000
min J_{train}	3900 ± 100	1.139 ± 0.161
min J_{val}	2720 ± 1064	1.018 ± 0.024
min R_{strong}	360 ± 152	4.146 ± 0.274

Table 12: Perforated plate: hierarchical diagnostics at the η_{48} -selected checkpoint. Ratios involving E_{FEM} are external validation quantities; $\mathcal{U}_{48,96,0.9}/\eta_{48}$ is the reference-free conditional bracket factor.

Quantity	Value
η_{24}/E_{FEM}	0.977 ± 0.002
η_{48}/E_{FEM}	0.993 ± 0.001
η_{96}/E_{FEM}	0.998 ± 0.000
$\delta_{48,96}/E_{\text{FEM}}$	0.093 ± 0.005
$\gamma_{24,48,96}$	0.517 ± 0.001
$\mathcal{U}_{48,96,0.9}/\eta_{48}$	1.023 ± 0.002
$\mathcal{U}_{48,96,0.9}/E_{\text{FEM}}$	1.016 ± 0.002

Both η_{48} and η_{96} select the FEM-reference oracle in all five runs (Table 11). The final iterate is 20.4% worse on average, validation Ritz selection remains close to the oracle, and the strong residual selects a much less accurate checkpoint.

We next assess the sharpness of the reference-free hierarchy at the selected checkpoint.

The auxiliary hierarchy is tightly calibrated on the FEM-reference scale (Table 12): η_m/E_{FEM} increases from 0.977 ± 0.002 at level 24 to 0.998 ± 0.000 at level 96. Independently of the FEM reference, the conditional $q = 0.9$ bracket has upper-to-lower factor 1.023 ± 0.002 ; the FEM comparison gives the external check $\mathcal{U}_{48,96,0.9}/E_{\text{FEM}} = 1.016 \pm 0.002$.

To assess the mechanical impact of checkpoint selection, we compare the η_{48} -selected output with the final iterate at the solution level.

The late-iterate degradation is also visible in the solution-level errors (Table 13): relative to the η_{48} -selected output, the final iterate increases the displacement, strain, von Mises stress, and sampled maximum-displacement errors by factors 1.405, 1.140, 1.206, and 1.275, respectively.

Table 13: Perforated plate: solution-level comparison with the refined FEM reference. Here $e_{u,L^2} = \|u_\theta - u_{\text{FEM}}^{\text{ref}}\|_{L^2} / \|u_{\text{FEM}}^{\text{ref}}\|_{L^2}$, $e_\varepsilon = \|\varepsilon(u_\theta - u_{\text{FEM}}^{\text{ref}})\|_{L^2} / \|\varepsilon(u_{\text{FEM}}^{\text{ref}})\|_{L^2}$, and $e_{\text{vm}} = \|\sigma_{\text{vm}}(u_\theta) - \sigma_{\text{vm}}(u_{\text{FEM}}^{\text{ref}})\|_{L^2} / \|\sigma_{\text{vm}}(u_{\text{FEM}}^{\text{ref}})\|_{L^2}$, where σ_{vm} denotes the plane-strain von Mises equivalent stress. The sampled maximum-displacement error is $e_{u,\text{max}} := \max_{x \in \mathcal{S}} \|u_\theta(x) - u_{\text{FEM}}^{\text{ref}}(x)\|_2$, where $\mathcal{S}$ is the union of the reference-mesh nodes and the degree-two triangle quadrature points; the final row reports final/selected ratios.

Selected output	E_{FEM}	e_{u,L^2}	e_ε	e_{vm}	sampled $e_{u,\text{max}}$
min η_{48}	$4.27 \times 10^{-3} \pm 2.65 \times 10^{-4}$	0.006 ± 0.001	0.085 ± 0.005	0.061 ± 0.002	$1.22 \times 10^{-3} \pm 1.50 \times 10^{-4}$
final	$5.15 \times 10^{-3} \pm 7.10 \times 10^{-4}$	0.008 ± 0.002	0.097 ± 0.011	0.073 ± 0.012	$1.56 \times 10^{-3} \pm 2.30 \times 10^{-4}$
final / min η_{48}	1.204	1.405	1.140	1.206	1.275

Table 14: Perforated plate: external FEM-solve cost and Riesz monitoring cost. FEM times are one-time reference solves used only for validation; Riesz times are mean costs per logged checkpoint, with overhead measured against the pure optimizer-update time between logs.

Quantity	DOFs	Time (s)	Overhead
FEM ref. 256	179882	53.39	–
FEM check 192	109608	34.47	–
η_{24}	2642	0.212 ± 0.055	0.93%
η_{48}	10536	0.959 ± 0.250	4.19%
η_{96}	42080	3.621 ± 0.677	15.84%

The operational η_{48} monitor adds 4.19% overhead in this benchmark (Table 14); the more expensive η_{96} reconstruction is used only for post-training qualification. Thus the reference-free selector operates at moderate cost without requiring the refined FEM solve in the selection pipeline.

Taken together, the manufactured benchmarks validate energy-scale calibration, refinement, and oracle-level selection, while the perforated-plate experiment supports the use of the Riesz monitor as a lightweight, reference-free post-training selector when no exact solution is available.

7 Discussion and outlook

This work developed a reference-free archive-level checkpoint-selection framework for admissible neural approximations of symmetric coercive variational problems. Conforming Riesz reconstruction converts the exact residual–energy geometry into a computable, training-independent monitor, making the logged energy oracle recoverable without the exact solution or a reference solve. Monitor refinement yields eventual logged-oracle recovery, while, under saturation, nested reconstructions provide finite-resolution near-oracle bounds and interval-separation certificates. Logging-resolution control further quantifies the loss relative to a prescribed finite comparison trajectory and provides trajectory-oracle certificates. Across the numerical benchmarks, the monitor approaches the reference energy error under refinement and recovers oracle-level checkpoints once sufficiently resolved. On the perforated plate, this behavior is confirmed against an independent FEM reference, with modest observed post-processing cost.

The key mechanism is preservation of the oracle–non-oracle ordering. Checkpointwise recovery alone does not ensure archive-level selection because finite-dimensional projection defects can reverse this ordering; finite-archive uniform recovery eventually restores it. This ordering issue becomes more fundamental beyond the symmetric coercive setting. Indeed, residual–error norm equivalence alone does not preserve the target-error ordering. Let $X = Y = \mathbb{R}^2$ with the Euclidean norm and

$$B = \text{diag}(1, 2), \quad e_i = (0, 1), \quad e_j = (3/2, 0).$$

Then

$$\|e\|_2 \leq \|Be\|_2 \leq 2\|e\|_2 \quad \forall e \in \mathbb{R}^2,$$

but

$$\|e_i\|_2 < \|e_j\|_2, \quad \|Be_i\|_2 > \|Be_j\|_2.$$

Thus, even exact norm equivalence does not in general preserve the ordering relevant to archive selection. In the present setting, by contrast,

$$\|R(u_{\theta_k})\|_{V'_{0,a}} = E_k,$$

so the continuous residual and energy-error orderings coincide; the ordering obstruction analyzed here is therefore introduced by the finite-dimensional reconstruction.

Natural extensions are saturation-free finite-resolution certification and archive-level selection beyond symmetric coercive formulations. Explicit quadrature-error control could further upgrade the trajectory-wide diagnostics to fully certified continuous-energy-norm bounds.

The framework concerns selection among candidates generated by the optimizer. Viewed from this archive-level perspective, a further natural direction is to carry the order-preservation principle upstream into the optimization procedure. We leave this direction for future work.

A Sensitivity to the prescribed saturation parameter

The prescribed saturation parameter q enters only the conditional upper estimate $\mathcal{U}_{m,M,q} = (\eta_m^2 + \delta_{m,M}^2 / (1 - q^2))^{1/2}$ and therefore does not affect monitor-based checkpoint selection. We assess the sensitivity of the finite-resolution certificates and upper estimates over the range $q \in \{0.70, 0.80, 0.90, 0.96\}$.

Table 15 reports the resulting interval-separation certification counts on κ_4 over five independent trajectories.

Table 15: Sensitivity of interval-separation certification on κ_4 . Each entry is the number of certified runs out of five.

q	$V_{24} \subset V_{48}$				$V_{48} \subset V_{96}$			
	$\mathcal{K}_{50}$	$\mathcal{K}_{100}$	$\mathcal{K}_{200}$	$\mathcal{K}_{400}$	$\mathcal{K}_{50}$	$\mathcal{K}_{100}$	$\mathcal{K}_{200}$	$\mathcal{K}_{400}$
0.70	2/5	5/5	4/5	5/5	3/5	5/5	4/5	5/5
0.80	1/5	4/5	4/5	5/5	3/5	5/5	4/5	5/5
0.90	0/5	2/5	4/5	4/5	2/5	5/5	4/5	5/5
0.96	0/5	0/5	2/5	4/5	0/5	3/5	4/5	5/5

The certification pattern remains stable over a broad range of q , particularly for $V_{48} \subset V_{96}$, with the expected gradual loss of certificates as the upper bounds become more conservative. Larger counts on coarser subarchives reflect greater checkpoint separation rather than improved trajectory coverage.

For the perforated plate, Table 16 reports the conditional upper estimate relative to the independent FEM-reference error at the η_{48} -selected checkpoint.

Table 16: Sensitivity of the conditional upper estimate on the perforated plate.

q	$\mathcal{U}_{48,96,q}/E_{\text{FEM}}$
0.70	1.002 ± 0.000
0.80	1.005 ± 0.001
0.90	1.016 ± 0.002
0.96	1.048 ± 0.005

The upper estimate remains close to the FEM-reference error throughout the tested range, reaching only 1.048 ± 0.005 at $q = 0.96$. Thus the qualitative conclusions remain stable over the tested range of q , with the expected increase in conservatism as q grows.

B Spatial diagnostics of the conforming Riesz reconstruction

For two manufactured stress tests, we compare the reference energy-error density with the spatial density of the conforming Riesz reconstruction. These comparisons provide a qualitative view of the dominant energetic regions captured by the reconstruction.

Figure 6 shows that the Riesz-projected density identifies the same dominant energetic region associated with the localized material transition as the reference density.

Figure 7 shows that both densities concentrate primarily near the reentrant corner while retaining nonzero contributions away from it, consistent with the observed need for global auxiliary refinement.

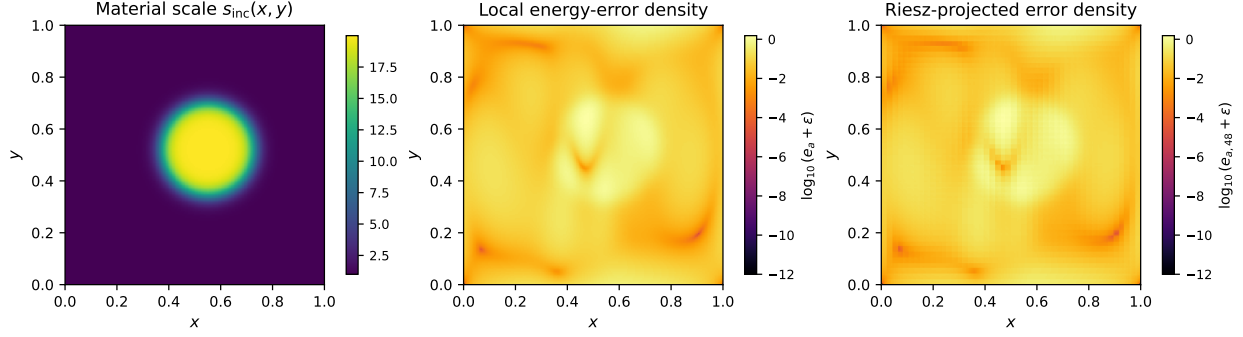


Figure 6: Spatial energy diagnostics for the smoothed high-contrast inclusion. Left: material scale $s_{\text{inc}}(x, y)$. Middle: reference energy-error density $e_a(x) = 2\mu|\varepsilon(u_\theta - u^*)|^2 + \lambda|\text{div}(u_\theta - u^*)|^2$. Right: conforming Riesz-projected density $e_{a,48}(x) = 2\mu|\varepsilon(z_{48})|^2 + \lambda|\text{div} z_{48}|^2$. Both densities are displayed as $\log_{10}(\cdot + \varepsilon)$.

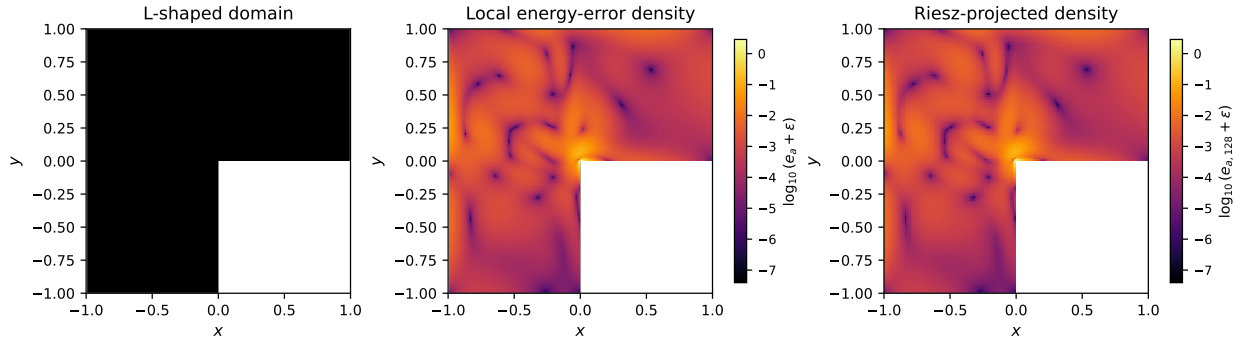


Figure 7: Spatial diagnostics for the L-shaped-domain stress test. Left: domain geometry. Middle: reference energy-error density $e_a(x) = |\nabla(u_\theta - u^*)(x)|^2$. Right: conforming Riesz-projected density $e_{a,128}(x) = |\nabla z_{128}(x)|^2$ on the 128×128 auxiliary space. Densities are shown as $\log_{10}(\cdot + \varepsilon)$.

C Discrete-hierarchy and quadrature-refinement audit

All audits below are post-training and use the same trained trajectories and saved checkpoints as the main experiments.

C.1 Monitor-hierarchy audit

For the scalar-diffusion and L-shaped benchmarks, we recompute the reported monitor diagnostics at the saved operationally selected checkpoints on the same auxiliary levels using the common finest-level Galerkin restrictions of Section 6. No monotonicity violation is observed. Increasing the quadrature order of the common Galerkin assembly from 3 to 5 changes the monitor values by at most 1.314×10^{-6} for scalar diffusion and 6.486×10^{-4} for the L-shaped problem.

C.2 Validation-oracle stability

As noted after Proposition 2, a quadrature approximation can perturb the ordering induced by an exact energy criterion. For the manufactured benchmarks, E_{quad} numerically defines the validation oracle, and we assess whether the reported ordering is stable under quadrature refinement. Without retraining, the manufactured-reference energies are reevaluated at tensor-product Gauss orders $Q = 3, 5, 7$. Table 17 reports the largest checkpointwise relative changes.

The reference-energy evaluations are stable under refinement. The largest sensitivity occurs for the singular L-shaped problem and decreases from $Q = 3 \rightarrow 5$ to $Q = 5 \rightarrow 7$. On the complete κ_4 archive, the oracle and runner-up remain unchanged in all five trajectories, while the smallest relative oracle-runner-up gap, 1.986×10^{-3} , remains well above the observed quadrature variations. The trajectory-oracle inclusion counts

Table 17: Manufactured-reference quadrature-refinement audit. The first three rows concern saved decision-relevant outputs; the κ_4 row uses the complete 205-checkpoint comparison archive.

Benchmark	Audited outputs	Maximum $Q = 3 \rightarrow 5$	Maximum $Q = 5 \rightarrow 7$
Scalar diffusion	20	1.494×10^{-7}	1.494×10^{-7}
L-shaped domain	5	2.932×10^{-3}	5.721×10^{-4}
Elasticity	16	1.043×10^{-7}	1.237×10^{-7}
κ_4 full archive	205	2.263×10^{-7}	4.759×10^{-8}

are also unchanged: $\mathcal{K}_{100} : 3/5$, $\mathcal{K}_{200} : 1/5$, and $\mathcal{K}_{400} : 0/5$. Thus the validation ordering underlying the reported κ_4 oracle and logging-resolution conclusions is insensitive to the tested quadrature refinements.

For the perforated plate, refining the independent FEM reference from level 192 to 256 likewise preserves the FEM-reference oracle in all five runs, with maximum relative checkpoint-error change 6.26×10^{-4} and rank correlation at least 0.999826.

Together, these audits support numerical stability of the reported selection and logging-resolution conclusions under the tested quadrature and FEM refinements.

Remark 8 (Continuous-energy oracle certification). The refinement audit above establishes numerical stability with respect to the tested discretizations. Let E_k denote the exact continuous energy error and $\tilde{E}_k$ its numerical approximation for k in a finite archive $\mathcal{K}$. If $\max_{k \in \mathcal{K}} |\tilde{E}_k - E_k| \leq \varepsilon$ and the unique numerical minimizer $\tilde{k} \in \mathcal{K}$ has oracle-competitor gap $\min_{j \in \mathcal{K} \setminus \{\tilde{k}\}} (\tilde{E}_j - \tilde{E}_{\tilde{k}}) > 2\varepsilon$, then $\tilde{k}$ is also the unique continuous-energy oracle, since $E_j - E_{\tilde{k}} \geq \tilde{E}_j - \tilde{E}_{\tilde{k}} - 2\varepsilon > 0$ for every $j \in \mathcal{K} \setminus \{\tilde{k}\}$. Thus certification of an E_{quad} -based validation oracle against the exact continuous energy would follow from a suitable uniform numerical-integration error bound; such bounds are left to future work.

References

- [1] Tim De Ryck and Siddhartha Mishra. “Numerical analysis of physics-informed neural networks and related models in physics-informed machine learning”. In: *Acta Numerica* 33 (2024), pp. 633–713. DOI: 10.1017/S0962492923000089.
- [2] Marius Zeinhofer, Rami Masri, and Kent-André Mardal. “A unified framework for the error analysis of physics-informed neural networks”. In: *IMA Journal of Numerical Analysis* 45.5 (2025), pp. 2988–3025. DOI: 10.1093/imanum/drae081.
- [3] Yeonjong Shin, Zhongqiang Zhang, and George Em Karniadakis. “Error estimates of residual minimization using neural networks for linear PDEs”. In: *Journal of Machine Learning for Modeling and Computing* 4.4 (2023), pp. 73–101. DOI: 10.1615/JMachLearnModelComput.2023050411.
- [4] Piotr Minakowski and Thomas Richter. “A priori and a posteriori error estimates for the Deep Ritz method applied to the Laplace and Stokes problem”. In: *Journal of Computational and Applied Mathematics* 421 (2023). See also the erratum in *J. Comput. Appl. Math.* 460 (2025), 116406, doi:10.1016/j.cam.2024.116406, p. 114845. DOI: 10.1016/j.cam.2022.114845.
- [5] Stefano Berrone, Claudio Canuto, and Moreno Pintore. “Solving PDEs by variational physics-informed neural networks: An a posteriori error analysis”. In: *Annali dell’Università di Ferrara* 68 (2022), pp. 575–595. DOI: 10.1007/s11565-022-00441-6.
- [6] Andrea Bonito, Ronald DeVore, Guergana Petrova, and Jonathan W. Siegel. “Convergence and error control of consistent PINNs for elliptic PDEs”. In: *IMA Journal of Numerical Analysis* 46.1 (2026), pp. 90–148. DOI: 10.1093/imanum/draf008.
- [7] Markus Bachmayr, Wolfgang Dahmen, and Mathias Oster. “Variationally correct neural residual regression for parametric PDEs: On the viability of controlled accuracy”. In: *IMA Journal of Numerical Analysis* (2025). DOI: 10.1093/imanum/draf073.
- [8] Sergio Rojas, Paweł Maczuga, Judit Muñoz-Matute, David Pardo, and Maciej Paszyński. “Robust variational physics-informed neural networks”. In: *Computer Methods in Applied Mechanics and Engineering* 425 (2024), p. 116904. DOI: 10.1016/j.cma.2024.116904.

- [9] Thomas Führer and Sergio Rojas. *A posteriori analysis of neural network approximations*. 2025. DOI: 10.48550/arXiv.2507.06017. arXiv: 2507.06017 [math.NA].
- [10] Lewin Ernst, Nikolaos Rekatsinas, and Karsten Urban. *A posteriori certification of PDE approximations with particular application to neural networks*. 2025. DOI: 10.48550/arXiv.2502.20336. arXiv: 2502.20336 [math.NA].
- [11] Mark Ainsworth and Justin Dong. “Galerkin neural networks: A framework for approximating variational equations with error control”. In: *SIAM Journal on Scientific Computing* 43.4 (2021), A2474–A2501. DOI: 10.1137/20M1366587.
- [12] Mingxing Weng, Zhiping Mao, and Jie Shen. “Deep collocation method: A framework for solving PDEs using neural networks with error control”. In: *SIAM Journal on Scientific Computing* 48.1 (2026), pp. C77–C102. DOI: 10.1137/25M1739753.
- [13] Mark Ainsworth and Justin Dong. “Extended Galerkin neural network approximation of singular variational problems with error control”. In: *SIAM Journal on Scientific Computing* 47.3 (2025), pp. C738–C768. DOI: 10.1137/24M1658279.
- [14] Philippe G. Ciarlet. *The Finite Element Method for Elliptic Problems*. Amsterdam: North-Holland, 1978.
- [15] Susanne C. Brenner and L. Ridgway Scott. *The Mathematical Theory of Finite Element Methods*. 3rd ed. New York: Springer, 2008. DOI: 10.1007/978-0-387-75934-0.
- [16] Randolph E. Bank and Alan Weiser. “Some a posteriori error estimators for elliptic partial differential equations”. In: *Mathematics of Computation* 44.170 (1985), pp. 283–301. DOI: 10.1090/S0025-5718-1985-0777265-X.
- [17] Mark Ainsworth and J. Tinsley Oden. *A Posteriori Error Estimation in Finite Element Analysis*. New York: Wiley, 2000. DOI: 10.1002/9781118032824.
- [18] Rüdiger Verfürth. *A Posteriori Error Estimation Techniques for Finite Element Methods*. Oxford: Oxford University Press, 2013. DOI: 10.1093/acprof:oso/9780199679423.001.0001.
- [19] Rüdiger Verfürth. *A Review of A Posteriori Error Estimation and Adaptive Mesh-Refinement Techniques*. Chichester: Wiley–Teubner, 1996.
- [20] Weinan E and Bing Yu. “The Deep Ritz method: A deep learning-based numerical algorithm for solving variational problems”. In: *Communications in Mathematics and Statistics* 6.1 (2018), pp. 1–12. DOI: 10.1007/s40304-018-0127-z.